BEST 'A' LEVEL BIOLOGY NOTES®

INHERITED CHANGE AND EVOLUTION

TOPICS:

1. MONOHYBRID INHERITANCE
2. DIHYBRID INHERITANCE
3. INCOMPLETE DOMINANCE
4. CODOMINANCE
5. CHI-SQUARED TEST
6. SEX LINKAGE

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8.5 TOPIC 5 INHERITED CHANGE AND EVOLUTION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (ATTITUDES, SKILLS AND KNOWLEDGE)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.5.1 Nature of Gene	• discuss the gene concept	- Gene as unit of inheritance	• Discussing the gene concept.	• ICT tools • Braille software/Jaws • Print media
8.5.2 Monohybrid and Dihybrid crosses	• use genetic diagrams to solve problems involving monohybrid and dihybrid crosses • use chi – squared test to test the significance of difference between observed and expected results	- Co-dominance - Sex linkage - Multiple alleles - Test crosses - Chi – squared test	• Performing genetic crosses. • Demonstrating genetic crosses using beads, seeds or pebbles. • Applying the chi-squared test to results obtained from the demonstrations.	• Print media • Seeds • Pebbles • Beads • Scientific calculator • Statistical tables

- **Heredity** – is the passing of characteristics from one generation to the next.
- **Genetics** is the study of heredity.

— **Inheritance**: Process by which characteristics are passed on from parent to progeny.
— **Variation**: Degree by which the progeny differs from its parents.

Keywords in genetics:
- **Gene** – A unit of heredity. A section of DNA sequence encoding a single protein.
- **Genome** – The entire set of genes in an organism.
- **Allele** – one of a number of alternative forms of a gene. Two genes for the same trait that occupy the same position on homologous chromosomes (like versions of a trait).
- **Locus** – the specific position of a gene on a chromosome, the two alleles of a gene are found at the same loci on the chromosome pairs.
- **Haploid (n)** – haploid cells carry a single set of chromosomes in the nucleus. These include sperm and egg cells. Produced through meiosis.
- **Diploid (2n)** – diploid cells carry two sets of chromosomes in the nucleus. Produced through mitosis.
- **Phenotype** – The physical appearance of an organism (observable characteristics of an organism which are as a result of genotype and environment)
- **Genotype** – the genetic makeup of an organism. (The alleles present within cells of an organism, for a particular trait or characteristic).
- **Dominant** – The allele of a gene that masks or suppresses the expression of an alternate allele. Indicated by a capital (uppercase) letter (e.g. T). Only a single allele is required for the characteristic to be expressed, that is, the allele is always expressed in the phenotype.
- **Recessive** – An allele that is masked by a dominant allele. Represented by a small (lowercase) letter (e.g. t). The characteristic is only expressed if there is no dominant allele present.
- **Homozygous** – Having two identical alleles for a particular characteristic (e.g. TT or tt).
- **Heterozygous** – Having two different alleles for a particular characteristic (e.g. Tt).

- **Homologous chromosomes** – a pair of chromosomes, one from the father and one from the mother. Each has the same genes in the same position, although the alleles of these genes may be different.
- **Linkage** is the phenomenon where genes for different characteristics, located at different loci on the same chromosome are linked.
- **Monogenic inheritance** – when a phenotype or trait is controlled by a single gene. For instance, cystic fibrosis where the individuals with doubly recessive phenotype are affected.
- **Monohybrid cross** – a genetic cross involving inheritance of a single pair of genes (one trait).
- **Dihybrid cross** – a genetic cross involving inheritance of two genes.
- **Punnett square** – a tool to do genetic crosses and predict the possible genotype and phenotype outcomes and probabilities.
- **Sex linkage** – expression of a gene whose locus is on the X chromosome. Expression of the allele depends on the gender of the individual as the gene is located on a sex chromosome, for instance, males are more likely to inherit an X-chromosome linked condition because they only have a single copy of the X chromosome. An example of sex linkage is haemophilia which is a recessive condition (hh).
- **Autosomal linkage** – genes which are located on the same chromosome (not the sex chromosomes) and tend to be expressed together in the offspring.
- **Codominance** – when both alleles are expressed in a heterozygote, that is, both alleles contribute towards the phenotype. Examples AB blood type. Alleles I^A and I^B are both dominant (i.e. codominant), so they will both be expressed to create the phenotype AB.
- **Multiple Alleles** – More than two alleles for a single gene. Example there are three alleles (multiple alleles) of blood types namely I^A I^B and i^o (or simply, A, B and O).

Gregor Mendel (1822-1884)

• German monk born in 1822.
• Lived in what is now the Czech Republic.
• Tended the garden at his monastery, conducted experiments with garden pea plants for 7 years.
• Went to the University of Vienna, where he studied botany and statistics and learned the Scientific Method.

- Applied statistics to the results of his experiments with the pea plant (*Pisum sativum*).
- In 1866 he published *Experiments in Plant Hybridization*, in which he established his three Principles of Inheritance:

1) Genes exist
2) Genes occur in pairs
3) One gene of each pair is present in the gametes
   - Unfortunately his work was largely ignored for 34 years, until 1900, when 3 independent botanists rediscovered it.
   - Now considered the "Father of Genetics".

Reasons for Selection of Garden Pea by Mendel:

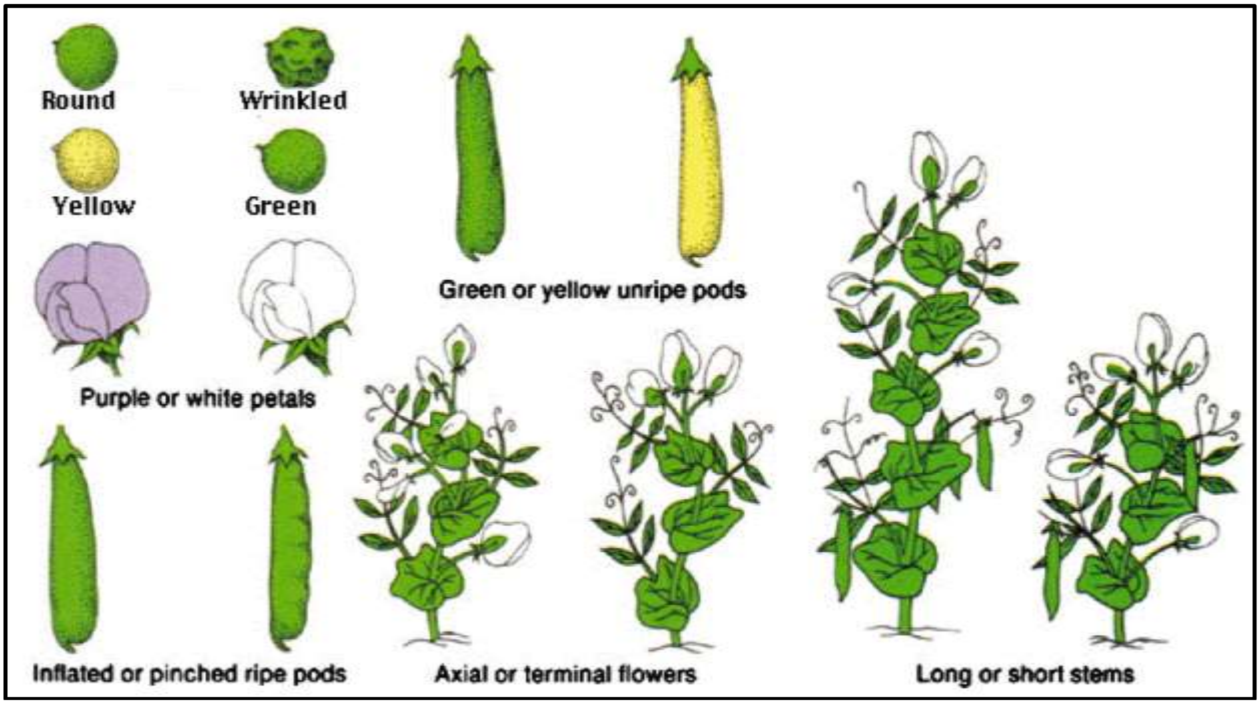
1. Garden pea is an annual plant and completes the life cycle within three or four months. Due to this short lifespan, he was able to take three generations in a year.
2. It is a small herbaceous plant that produces many seeds and so he could grow thousands of pea plants in a small plot behind the church.
3. It is naturally self-pollinating and was available in the form of many varieties with contrasting characters. There were no intermediate characters.
4. Flowers are large enough for easy emasculation required for artificial cross and produce fertile offspring.

The Reason of Success of Mendel's Experiment:

1. Mendel studied the inheritance of one character at a time whereas earlier scientists had considered the organism as a whole. Initially, Mendel considered the inheritance of one trait only (Monohybrid). Then he studied two traits together (dihybrid) and then three (Trihybrid).
2. He started with pure line i.e. true breeding. He maintained a complete statistical record by counting an actual number of offspring.
3. He carried out experiments up to the second and third generations.
4. He conducted ample crosses and reciprocal crosses to eliminate chance.
5. He dealt with a large sample size.

What is a character? What is a trait?

- ***Character*** = Detectable inheritable feature of an organism
- ***Trait*** = Variant of an inheritable character
- A trait is a character that can vary from one individual to the next (e.g. flower color)
- Mendel chose characters in pea plants to study that differed in a relatively clear-cut manner. He chose seven characters, each of which occurred in two alternative forms or traits:
  1. Flower color (purple or white)
  2. Flower position (axial or terminal)
  3. Seed color (yellow or green)
  4. Seed shape (round or wrinkled)
  5. Pod shape (inflated or constricted)
  6. Pod color (green or yellow)
  7. Stem length (tall or dwarf/short)
- He was lucky that each trait happened to be located on different chromosomes (people didn't know about chromosomes back then)



PICTURE: Seven pea characteristics studied by Mendel.

- Mendel started by looking at a **single trait** and performed **single trait crosses** (a.k.a. **monohybrid crosses**).
- Later he performed **two trait crosses** (a.k.a. **dihybrid crosses**).
- While the **monohybrid cross** would yield **3:1 ratio of the phenotypes**, the **dihybrid cross** yielded **a 9:3:3:1 ratio** of all the combinations of each phenotype.

You need to be familiar with these terms that will be used in the following section:

- **Emasculation =** removal of anthers/stamens before the formation of pollen grains.
- **True (pure) breeding** = offspring always have same trait as parent when the parents are self-fertilised.
- **Self-pollination** = plant fertilizes itself.
- **Cross-pollination** = one plant fertilizes another.
- **P generation** = Parent generation.
- **P_1 generation** = Parent generation that produces the F_1 generation
- **F_1 generation** = 1st generation offspring ("filius" is Latin for "offspring"); offspring of P generation.
- **F_2 generation** = 2nd generation offspring (offspring of F_1 generation).

MENDEL'S MONOHYBRID CROSS-Mendel's experiments with single trait crosses.

- **Monohybrid cross** a.k.a. single trait cross– a genetic cross involving inheritance of a single pair of genes (one trait).

❖ **Mendel's first set of experiments:** He began with flower color.
  - Mendel **cross-pollinated purple- and white-flowered parent plants.**
- **Mendel's procedure was as follows:**

Step – 1: Selection of parents and obtaining Pure lines:

- First, he created **true breeding (pure breeding)** pea plants in terms of **flower colour** by selfing (self-pollinating/self-fertilising) them for three generations. Thus pure breeding purple flower and white flower plants were obtained.
- The pure line (pure breeding) plants are **homozygous** for a given trait e.g. in terms of flower colour, female plant was pure purple colour and male plant was pure white flower.

Step – 2: Emasculation, Dusting and Raising F1 Generation (Hybridization):

Emasculation:

- Emasculation is the removal of anthers/stamens before the formation of pollen grains (anthesis).
- This was done in the bud condition. The bud was carefully opened and all anthers/stamens were removed carefully.
- The stigma was then protected against any foreign pollen with the help of a muslin bag.
- Emasculation ensured that the plants did not self-pollinate but cross-pollinated during the experiment.

Dusting and Raising F1 Generation:

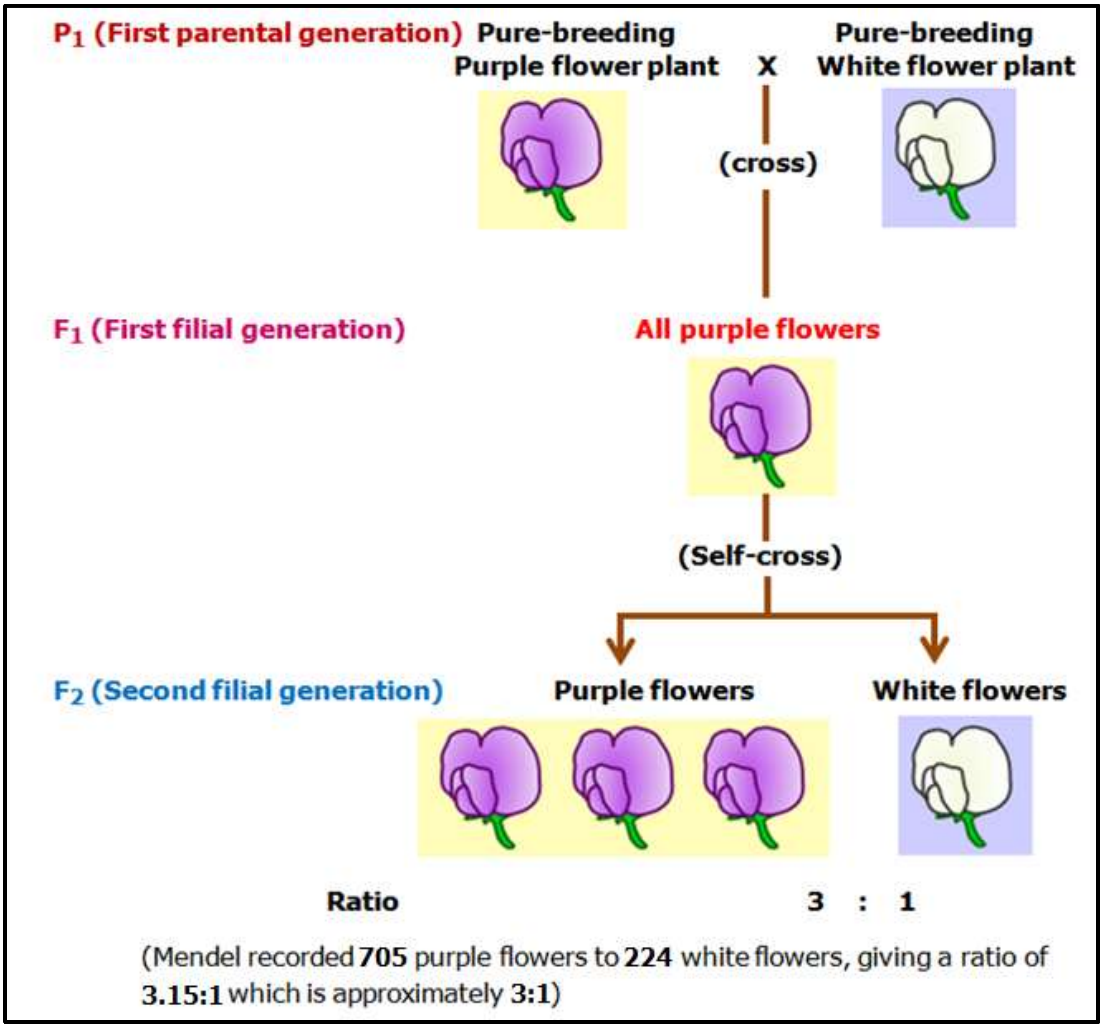
- The pollen grains from the selected male flowers were dusted on the stigma of an emasculated female flower using an artist's paint brush.
- The cross-pollinated flowers were enclosed in separate bags (bagging) to avoid further deposition of pollens from another source.
- During the pollination, it was assured that the pollen is mature and the stigma is receptive.
- This is an artificial cross.
- Mendel crossed many flowers, collected seeds and raised F1 generation.
- **All plants of F1 generation had purple flowers.**
- It seemed that the white colour had been lost or had disappeared.
- ***Hypothesis:*** Mendel hypothesized that if the inheritable factor for white flowers had been lost, then a cross between F1 plants should produce only purple-flowered plants.
- (A *hypothesis* is an intelligent guess that is not yet verified experimentally. To verify this hypothesis, Mendel carried out **step 3** as explained below).

Step – 3: Selfing of F1 hybrids to produce F2 Generation:

- Mendel allowed the F1 plants to self-pollinate. He collected the seeds and planted them to obtain the F2 generation.
- ***Results:*** There were 705 purple-flowered and 224 white-flowered plants in the F2 generation— a ratio of 3.15:1 which is approximately **3:1**. The inheritable factor for white flowers was not lost, so the hypothesis was rejected.

The **3:1** ratio became known as the **monohybrid ratio**.

➢ Mendel's results are shown in the diagram that follows:



❖ **Mendel's observations:**

1. After cross pollination (hybridization), the resulting F_1 generation:

- all **physically resembled <u>one</u> of the parents** (say **all** purple flowers, no white flowers).
- had **no mixed/intermediate colours** e.g. pink flowers which is a mixture of purple and white. This disproved the widely-held **blending theories of inheritance** that characteristics gradually mixed over time.

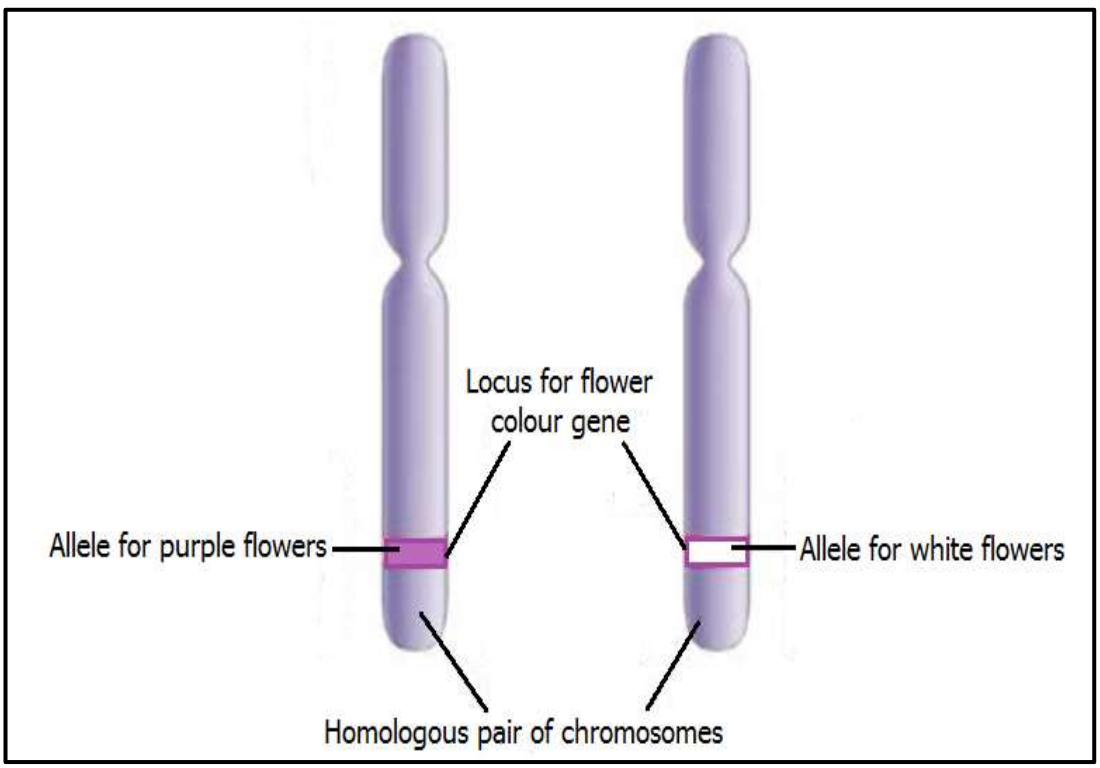
2. When the 2 parents from F_1 were self-pollinated the F_2 generation was obtained with the following observations:

- The **trait that was hidden (unexpressed) in F_1** (e.g. white flowers) was observed in some F_2 progeny.
- **Both traits**, e.g. purple flowers and white flowers, were **expressed in F_2 in the ratio 3:1** (i.e. ¾ to ¼ or 75% to 25%).
- The trait that was hidden in the F_1 generation and revived/expressed in F_2 was the least common of the two traits i.e. **the hidden trait of F_1 was expressed in ¼ or 25% of the F2** progeny.

❖ **<u>Mendel's conclusions</u> and today's understanding of genes and alleles:**

1. A characteristic can be hidden (**disappear**) for a generation, but then expressed (**reappear**) the following generation, looking exactly the same. So **a characteristic can be present but hidden**.
2. Mendel proposed that ***'something'*** is being passed on **unchanged** from generation to generation. He called these things **'factors'** (we now call the factors **genes**).
3. **'Factors'** contain and carry **hereditary information**.
4. We now know that **alleles** are **slightly different forms/versions of the same gene ('factor')**.
5. We now also know that the **two alleles code for a pair of two contrasting traits** (e.g., tall and dwarf).
6. One form of a characteristic can **mask** the other. The two forms are called **dominant** and **recessive** respectively. That is why one characteristic was hidden in F_1 and then reappeared in F_2. It was masked, i.e. the **recessive allele** was **masked** by the **dominant allele**.

❖ Mendel's **'factors'** are now called **genes** and the **two alternative forms of the gene** are called **alleles**.
- So in the example above we would say that there is a **gene for flower colour** and **its two alleles are "purple" and "white".** (The dominant allele is now represented by a capital letter, e.g. **P** for "purple", and the recessive allele by a small letter, e.g. **p** for "white").
- **One allele** comes from **each parent**, and the **two alleles** are found on the **same position (**or **locus)** on the **homologous chromosomes**.



- With two alleles there are three possible combinations of alleles (or genotypes) and two possible appearances (or phenotypes):

Genotype	Name	Phenotype
PP	homozygous dominant	Purple
pp	homozygous recessive	white
Pp	heterozygous	Purple

- Mendel referred to traits that showed up in F_1 generation as **dominant** and those that didn't show up in the F_1 generation as **recessive.**
- **Mendel also performed monohybrid crosses with the other six pea characteristics and always got consistent results each time with a ratio of 3 dominant : 1 recessive in the F_2 generation (see table below).**

TABLE 8.5: Results of Mendel's monohybrid crosses for seven characters in pea plants

Character	Parental cross: Dominant vs Recessive trait		F_1 phenotypes	F_2 phenotypes		Ratio
	Dominant trait	Recessive trait		Dominant form	Recessive form	
Flower color	Purple	White	All purple	705	224	3.15:1
Seed color	Yellow	Green	All yellow	6022	2001	3.01:1
Seed shape	Round	Wrinkled	All round	5474	1850	2.96:1
Pod color	Green	Yellow	All green	428	152	2.82:1
Pod shape	Inflated	Constricted	All inflated	882	299	2.95:1
Flower position	Axial	Top	All axial	651	207	3.14:1
Plant height	Tall	Dwarf	All tall	787	277	2.84:1

- From the **observations on monohybrid crosses**, Mendel came up with **two laws of inheritance** namely **The Law (Principle) of Dominance** and **The Law (Principle) of Segregation**.
- When he later carried out **dihybrid crosses** (two trait crosses) **Mendel added a third law** called **The Law (Principle) of Assortment**.
- So **altogether there are three Laws (Principles) of inheritance** that Mendel formulated from his experiments with pea plants.

❖ These laws do not apply to pea plants only but to most organism including human beings.

Mendel's Three Laws of Inheritance

- Based on his experiments, Mendel proposed three laws or principles of inheritance:
    1) Law (Principle) of Dominance
    2) Law (Principle) of Segregation
    3) Law (Principle) of Independent Assortment.
- Law of dominance and law of segregation are based on monohybrid cross while law of independent assortment is based on dihybrid cross.

Law of Dominance

- According to this law, characteristics are controlled by discrete units called factors (genes), which occur in pairs (as alleles) with one member of the pair dominating over the other.
- **Generally stated as: one allele is dominant over another.**
- An organism with at least one dominant allele will have the phenotype of the dominant allele.
- This law explains why only one of the parental characteristics was expressed in F_1 generation and both were expressed in F_2 generation.
- Dominant allele is always expressed, even if combined with recessive allele:
    - written as uppercase letter of the trait
    - e.g., tall = dominant: TT, or Tt.
    - The dominant phenotype will occur when the genotype contains either 2 dominant alleles (homozygous dominant e.g. TT) or one dominant and one recessive (heterozygous e.g. Tt)
- Recessive allele expressed only if dominant allele is not present:
    - written as a lowercase letter of the dominant trait
    - e.g., dwarf = recessive: tt
    - The recessive phenotype will only appear when the genotype contains 2 recessive alleles. This is referred to as **homozygous recessive** e.g. tt.

Law of Segregation

- This law states that the two alleles of a pair segregate or separate during gamete formation such that a gamete receives only one of the two factors.
- Generally stated as: **individuals carry two alleles for each gene, these alleles segregate in meiosis during production of sex cells.**

In homozygous parents, all gametes produced are similar; while in heterozygous parents, two different kinds of gametes are produced in equal proportions.

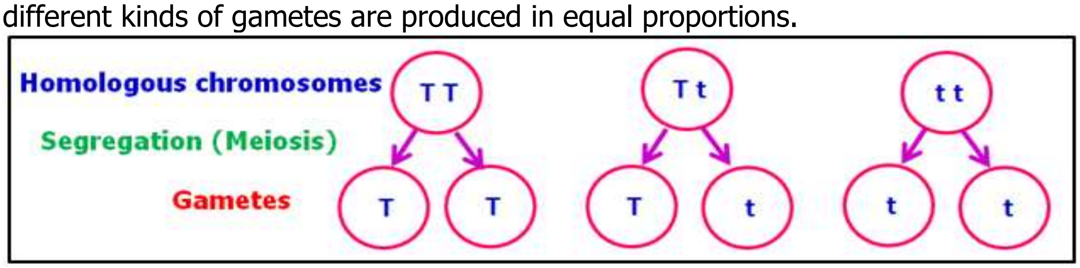


Diagram illustrating the law or principle of segregation.

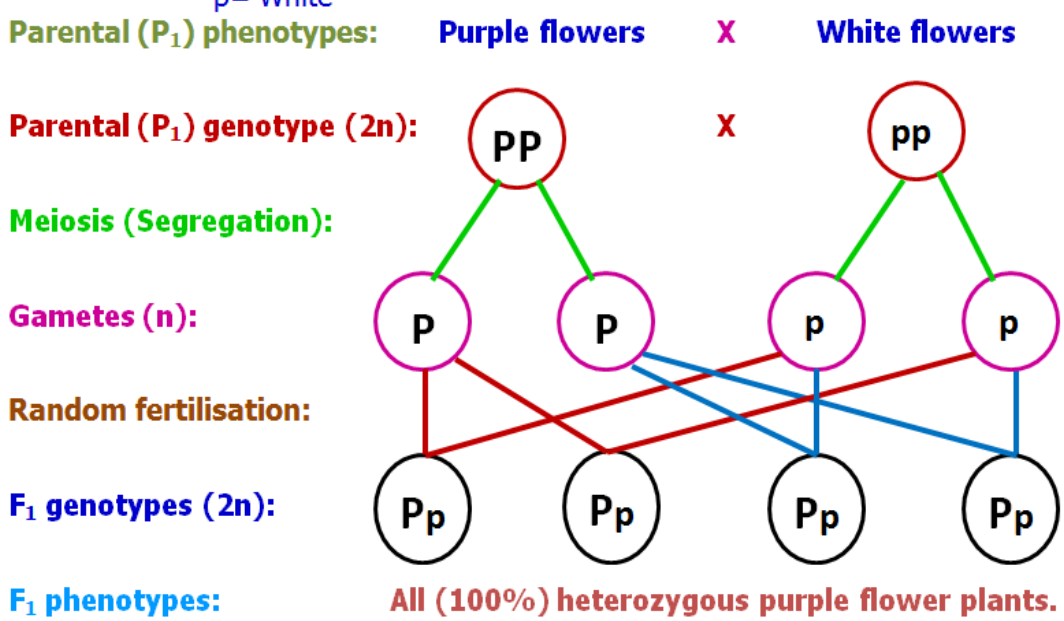
GENETIC CROSS DIAGRAMS

- **Mendel's observations can be analysed using genetic cross diagrams and Punnett squares.**

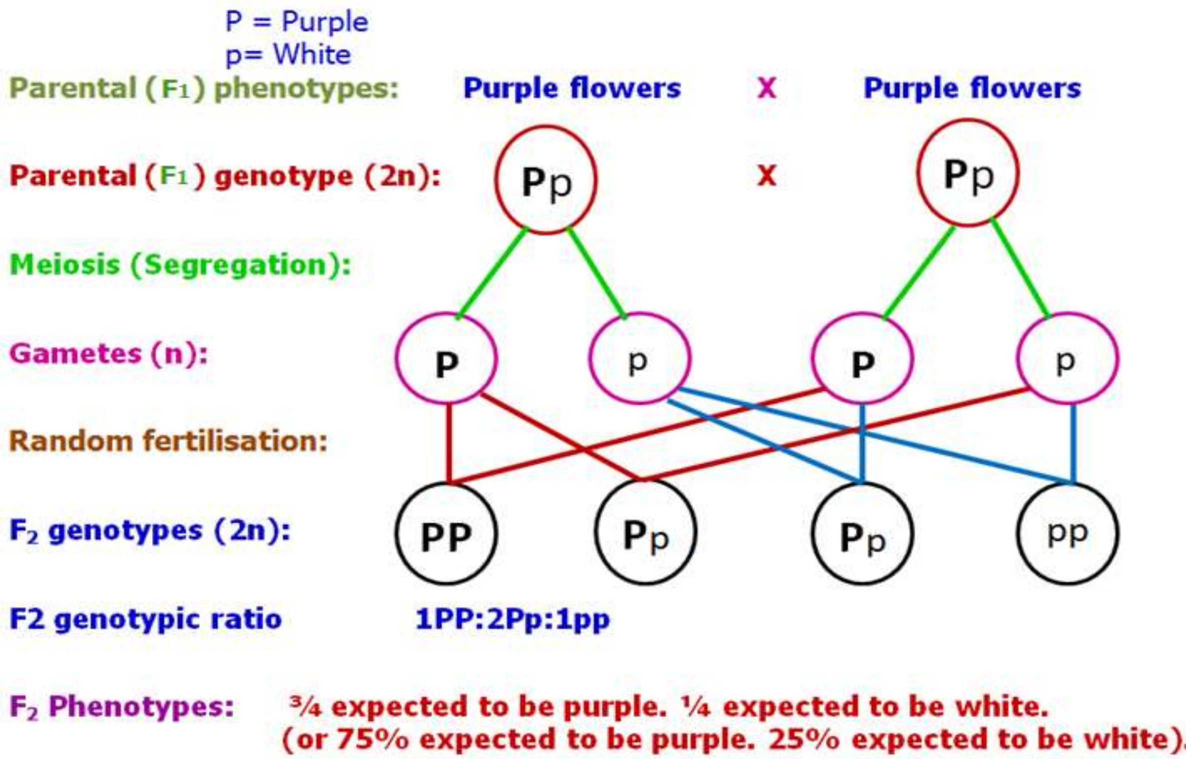
Genetic cross diagram

Mendel crossed pure breeding lines (homozygous parents) of peas. In his first experiment Mendel crossed a homozygous purple flower plant with a homozygous white flower plant to get the F_1 generation. He further crossed the F_1 offspring to get the F_2 generation. Purple flowers are dominant over white flowers in pea plants. The possible outcomes of F_1 and F_2 offspring can be predicted using a genetic cross diagram as shown below:

P = Purple
p= White



- **Conclusion:**
  1. **Principle of Dominance: some alleles are dominant and others are recessive.** Purple (P) is dominant over white (p). White (p) is recessive.
  2. **Principle of Segregation: Two alleles for a trait separate during meiosis and each gamete receives only one allele.** The alleles are on separate **homologous chromosomes.** P and p separated (segregated) during meiosis, and each gamete received only P or p.



❖ The F_1 generation were self-pollinated. The genetic cross diagram is as follows:

Punnett Square

- Named after Reginald C. Punnett, who devised the approach.
- The **Punnett square** is a square grid used to:
  1. predict the genotypes of a particular cross or breeding experiment.
  2. determine the probability of an offspring having a particular genotype.
- For the first examples, we will only be testing the **complete dominance** condition (where one allele completely dominates over the other).

Rules for predicting monohybrid crosses using a Punnett square:

3. Determine parent genotypes (capital letter for dominant, small letter for recessive).
4. Segregate alleles and place alleles of each parent on top and side of four-squared grid (mom's on one side and dad's on the other).
5. Combine parent alleles inside boxes (letters inside boxes show POSSIBLE genotypes of offspring- not ACTUAL).
6. Determine phenotypic ratio and possible genotypes in the following format:
   - Fraction, probability statement, phenotype (genotype)
   - Ex. ¾ cbetb (Could Be Expected To Be) purple (PP, Pp), ¼ cbetb white (pp).
   - MUST BE LIKE THIS EVERY TIME!!!!

Examples of monohybrid crosses

1. Mendel crossed pure breeding lines (homozygous parents) of peas. Purple flowers are dominant over white flowers in pea plants. In his first experiment Mendel crossed a homozygous purple flower plant with a homozygous white flower plant to get the F_1

generation. He further crossed the F_1 offspring to get the F_2 generation. Use Punnett squares to predict the possible outcomes of F_1 and F_2 offspring.

Answer

P = Purple
p= White

P_1 phenotypes: Purple flowers X White flowers
P_1 genotypes (2n): PP X pp
Meiosis (segregation):
Gametes (n): P P p p

F_1 genotypes and phenotypes:

White flower parent

Purple flower parent	p	p
P	Pp	Pp
P	Pp	Pp

F_1 Phenotypes: All (100%) expected to be purple.

F_1 genotypes: All (100%) expected to be Pp (Heterozygotes/Hybrids).

- **Conclusion: Principle of Dominance:** some alleles are dominant and others are recessive. Purple (P) is dominant over white (p). White (p) is recessive.

F1 offspring (progeny) were self-pollinated:

Parental (F_1) phenotypes: Purple flowers X Purple flowers
Parental (F_1) genotypes (2n): Pp X Pp
Meiosis (Segregation):
Gametes (n): P p P p

F_2 genotypes and phenotypes:

Purple flower plant

Purple flower plant	P	p
P	PP	Pp
p	Pp	pp

F_2 Phenotypes: ¾ expected to be purple. ¼ expected to be white.
(or75% expected to be purple. 25% expected to be white).

F_2 phenotypic ratio: 3 purple: 1 white.

F_2 genotypes: ¼ expected to be PP, ½ expected to be Pp and ¼ expected to be pp.

F_2 genotypic ratio 1PP:2Pp:1pp.

2. In humans, having **dimples is dominant** to not having dimples. Predict the genotypic and phenotypic ratios of a cross between a man heterozygous for dimples and a woman without dimples.

D = dimples
d = no dimples

Parental phenotypes:	Dimples	X	No dimples
Parental genotypes (2n):	Dd	X	dd
Meiosis (segregation):			
Gametes (n):	D d		d d

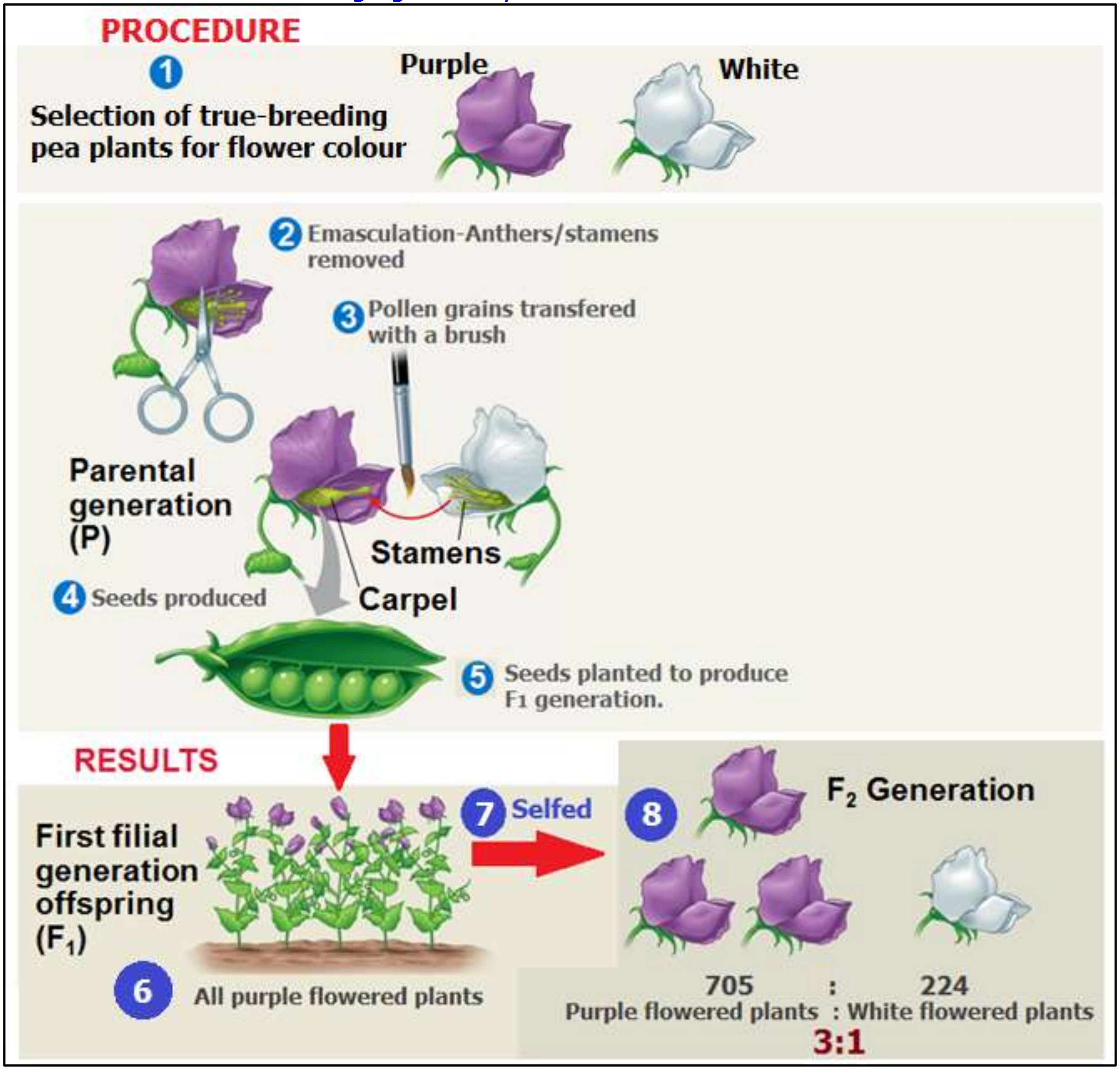
Offspring genotypes and phenotypes:

Mother ♀

Father ♂	d	d
D	Dd	Dd
d	dd	dd

Offspring Phenotypes: ½ chance of having dimples. ½ chance of having no dimples.
(or 50% chance of having dimples. 50% chance of having no dimples).
Offspring phenotypic ratio: 1 Dimples: 1 No dimples.
Offspring genotypes: ½ chance of being Dd and ½ chance of being dd.
Offspring genotypic ratio 1 Dd : 1 dd.

TASK 1: The following diagram shows the procedure and results of Mendel's Monohybrid experiment with flower colour of the pea plant. Describe stages 1 to 8 and use Punnett squares to show F_1 and F_2 generation results. Make reference to the Law of dominance and the Law of segregation in your answer.



TASK 2: By making reference to **Table 8.5**, use genetic cross diagrams and Punnett squares to predict the possible genotype and phenotype outcomes and probabilities of each and every of the seven characteristics studied by Mendel in his monohybrid cross experiments.

Activity: To demonstrate Monohybrid inheritance of Corn Colour in an F_2 Population

A corn cob contains hundreds of kernels. Each kernel is a seed that represents an individual organism. In the cob, we can easily see kernel color as a phenotype.

1. Retrieve an F_2 corn cob.
2. Count a total of 100 kernels.

   a) Tally the number of Yellow Kernels within that 100 (in the dried state, anything yellow or honey-colored counts as yellow).
   b) Tally the number of Purple Kernels within that 100 (in the dried state, purple colored kernels may appear brown).
   c) Ignore any speckled kernels that may have yellow and purple within them

3. Compare numbers with the class as a whole.
4. From the numbers:

   a) What is the ratio of Purple kernels to Yellow kernels?
   b) Is there a dominant color? ____
   c) Which is dominant, if there is? _____________________________________
   d) Which colour is recessive, if there is? ________________________________

e) Create a Punnett square to illustrate the expected number of each color in a simple dominant: recessive paradigm.

Typical results

Purple kernels		Yellow kernels	
Tally	Number	Tally	Number
𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 IIII	74	𝍸 𝍸 𝍸 𝍸 𝍸 I	26

a) What is the ratio of purple kernels to yellow kernels?
   74 purple kernels: 26 yellow kernel = 2.85: 1 = 3:1.
b) What name is given to this ratio? Monohybrid ratio.
c) Is there a dominant color? Yes.
d) Which is dominant, if there is? Purple.
e) Which colour is recessive, if there is? Yellow.
f) Create a Punnett square to illustrate the expected number of each color in a simple dominant: recessive paradigm.

<u>F1 offspring were self-pollinated to give the F2 generation</u>

P = Purple kernels
p = Yellow kernels.

Parental (F_1) phenotypes: Purple kernels X Purple kernels
Parental (F_1) genotypes (2n): Pp X Pp
Meiosis (Segregation):
Gametes (n): P p P p

F_2 genotypes and phenotypes:

Purple kernels	P	p
P	PP	Pp
p	Pp	pp

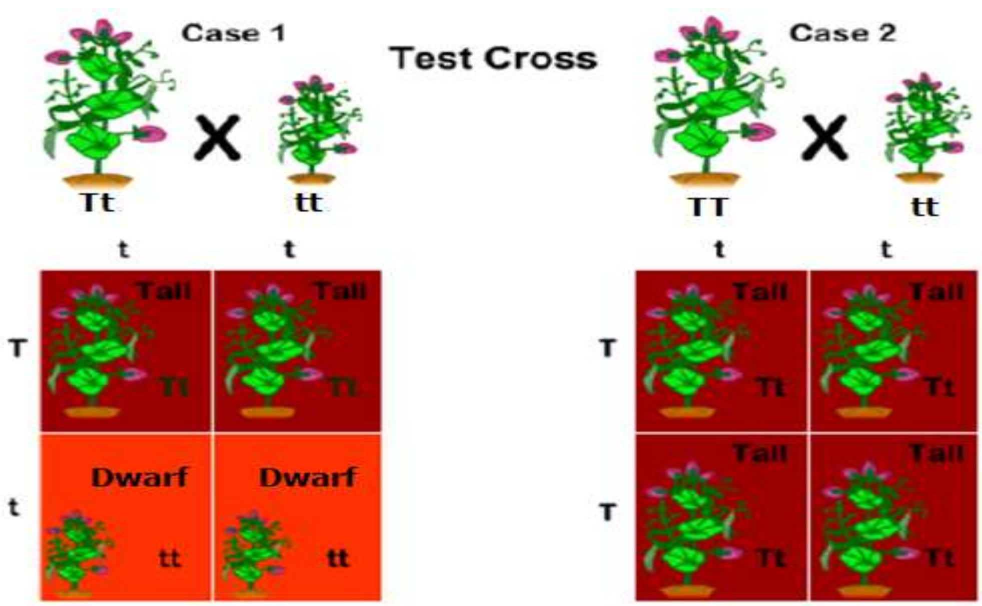
F_2 Phenotypes: ¾ expected to be purple. ¼ expected to be yellow.
(or75% expected to be purple. 25% expected to be yellow).
F_2 phenotypic ratio: 3 purple: 1 yellow.
F_2 genotypes: ¼ expected to be PP, ½ expected to be Pp and ¼ expected to be pp.
F_2 genotypic ratio 1PP:2Pp:1pp.

TEST CROSS

- **DEFINITION:** A test cross is the cross of any individual to a homozygous recessive parent; used to determine if the individual is homozygous dominant or heterozygous
- If the **phenotype of a parent** in a breeding experiment shows the **recessive trait** then it is pretty obvious that their **genotype is homozygous recessive** as it is the only genotype that will give that phenotype.
- However, if a **parent's phenotype shows the dominant trait** it means their **genotype** could either be **homozygous dominant** or **heterozygous**.
- Therefore a **test cross** is preformed **to find out** whether the **parent** is **homozygous dominant** or **heterozygous**.
- **How is this done?** The **parent** is **crossed** with the **'tester'** who is **homozygous recessive**.
  - If **all offspring** are of **one variety** then it means **the parent is homozygous dominant**.
  - If we obtain **offspring of both varieties** (in the **ratio 1:1**) then it means **the parent is heterozygous.**
- A test cross involving a **single trait** is called a **monohybrid test cross** and gives a **phenotypic ratio** of **1:1** as explained in the examples below.
- A test cross involving **two traits** is called a **dihybrid test cross** and it gives a **phenotypic ratio of 1:1:1:1**. We will look at this later in this document under the topic called 'Dihybrid cross'.

Examples of monohybrid test crosses

1. A test cross determines whether the dominant character is coming from homozygous dominant genotype or heterozygous genotype. (e.g., tallness in pea plants coming from TT or Tt).



- When TT is crossed with tt, we obtain all Tt (tall) individuals in the progeny. Whereas when Tt is crossed with tt, we obtain Tt (tall) and tt (dwarf) individuals in the progeny.

- Therefore, if tallness is coming from TT, then we obtain all tall progenies in a test cross. We obtain both tall and dwarf varieties in the ratio 1:1 in a test cross, if tallness is coming from Tt.

2. The genotypes and phenotypes of a tomato plant are as follows:

Genotype	Phenotype
AA	purple stem
Aa	purple stem
aa	green stem

A purple stem tomato plant might have the genotype **AA** or **Aa**; to find out its genotype, it could be crossed with a green-stemmed tomato plant **aa.**

If the purple-stemmed tomato plant's genotype is AA:

Parental phenotypes	purple	green
Parental genotypes	AA	aa
Gametes	A	a
Offspring **genotypes**	all Aa	
Offspring phenotypes	purple	

If its genotype is Aa:

Parental phenotypes	purple	green
Parental genotypes	Aa	aa
Gametes	A or a	a
Offspring **genotypes**	Aa	aa
Offspring phenotypes	purple	green
Phenotypic ratio	1 purple : 1 green	

- So from the colour of the offspring, it is possible to tell the genotype of the purple parent.
  - All offspring coming out purple means the unknown parent is homozygous dominant (AA in this case).
  - Appearance of both purple and green offspring (in the ratio 1:1) means the unknown parent is heterozygous (Aa in this case).

<u>DIHYBRID CROSSES</u> (**Two factors cross** – inheritance of two genes at once)

- In an organism, there are many characters and each character is controlled by respective alleles. To study whether one pair of alleles affects or influences the inheritance pattern of a pair of other alleles, Mendel performed dihybrid cross experiments. In a **dihybrid cross**, he considered **two traits simultaneously**.

- From the results of the dihybrid crosses he then proposed the **third law of inheritance** called the **Law (principle) of independent assortment**.
- A cross between two pure (homozygous) parents in which the inheritance pattern of two contrasting characters is studied is called a dihybrid cross. It is a cross between two pure (true-breeding) parents differing in two pairs of contrasting characters.
- Mendel studied the inheritance of round and wrinkled characters of seed coat along with the yellow and green colours of seeds. He found that a cross between round yellow and wrinkled green seeds (P1) produced only round yellow seeds in the F1 generation, but in F2 generation seeds of four phenotypes were observed. Two of these phenotypes were similar to the parental combinations (yellow round and green wrinkled), while the other two were new combinations (yellow wrinkled and green round).

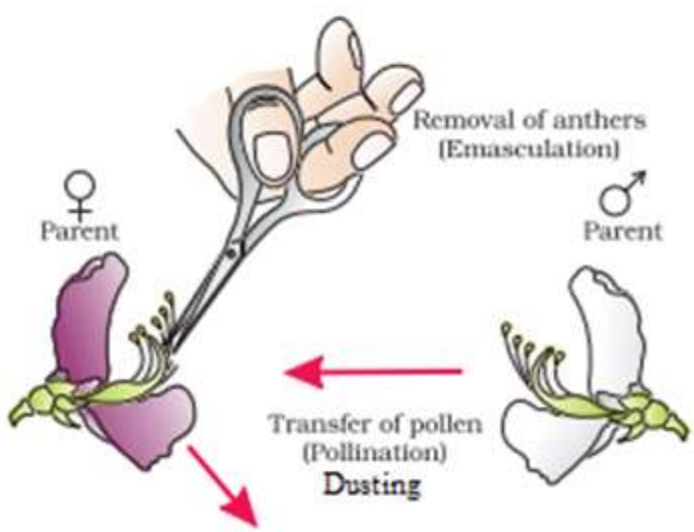
The Procedure of Dihybrid Cross Experiment

Step – 1: Selection of parents and obtaining Pure lines:

- For dihybrid cross, Mendel selected pea plant having round and yellow seeds (RRYY) as the female parent and pea plant having wrinkled and green (rryy) seeds as the male parent. He obtained pure line by selfing (self-pollinating) these plants for three generations. He confirmed that pea plant having round and yellow seeds are producing round and yellow seeds and pea plant having wrinkled and green seeds are producing wrinkled and green seeds.

Step – 2: Emasculation, Dusting and Raising F1 Generation:

Emasculation:
Emasculation is a process of removal of stamens before the formation of pollen grains. This is done in the bud condition. The bud is carefully opened and all stamens are removed carefully. Removal of stamens ensures that the plants do not self-pollinate but cross-pollinate during the experiment.



Dusting and Raising F1 Generation:

- The pollens from the selected male flowers are dusted onto the stigma of the emasculated female flower using a small paint brush. This is an artificial cross. Mendel crossed many flowers, collected seeds and raised F1 generation. The female plant produces gametes with genes RY while male plants produced gametes with genes ry.
- Round and Yellow are dominant alleles, hence all F1 Generation had round and yellow seeds. All the plants produced in F1 generation with yellow and round seeds (RrYy), which are heterozygous for both the alleles and are called dihybrid.

Punnett Square for F1 Generation:

♀ \ ♂	ry	ry
RY	RrYy	RrYy
RY	RrYy	RrYy

F_1 genotypes: All RrYy
F_1 phenotypes: All heterozygous round yellow seeds

Step – 3: Selfing of F_1 hybrids to Produce F2 Generation:

Mendel allowed natural pollination in each F_1 hybrid; collected seeds separately and planted them to obtain the F_2 generation.

Punnett Square for F2 Generation:

♀ \ ♂	RY	rY	Ry	ry
RY	RRYY	RrYY	RRYy	RrYy
rY	RrYY	rrYY	RrYy	rrYy
Ry	RRYy	RrYy	RRyy	Rryy
ry	RrYy	rrYy	Rryy	rryy

F_2 genotypes: (As listed in each square).
F_2 phenotypes: 9Round yellow:3Round green:3Wrinkled yellow:1Wrinkled green

Observations of the Dihybrid Cross Experiment:

Expectations:

Mendel expected the ratio of yellow and round seeds to green and wrinkled seeds to be 3:1.

Outcome:

He found seeds of four types, round yellow, wrinkled yellow, round green and wrinkled green in the ratio 9:3:3:1.

Out of these four types, two were parental combinations. Viz. yellow round and green wrinkled and two were new combinations like yellow wrinkled and green round.

In all Mendelian dihybrid crosses the ratio in which four different phenotypes occurred was **9:3:3:1**. This ratio is called the **dihybrid ratio**.

Phenotypic ratio i.e. the ratio of the yellow round, yellow wrinkled, green round and green wrinkled in the ratio 9:3:3:1.

What is Mendel's Law of Independent Assortment?

- Genes for two different traits are inherited independently
- There is no connection between them (e.g., seed colour and seed texture in pea plants).

Mathematical Explanation of Mendel's Law of Independent Assortment:

The meaning of the word assortment is 'randomly and freely'. Thus **probability theory is applicable to the dihybrid cross** experiment. By the basic principle of probability, "Probability of two independent events occurring simultaneously is a product of their individual probabilities".

The probability of the first trait is 3:1 while that of the second trait is also 3:1. Thus the dihybrid ratio should be **(3:1) x (3:1)** = **3 x 3 : 3 x 1 : 1 x 3 : 1 x 1** = **9:3:3:1** and Genotypic ratio RRYY: RrYY: RRYy: RrYy: rrYy: rrYy: RRyy: Rryy: rryy is 1:2:2:4:1:2:1:2:1.

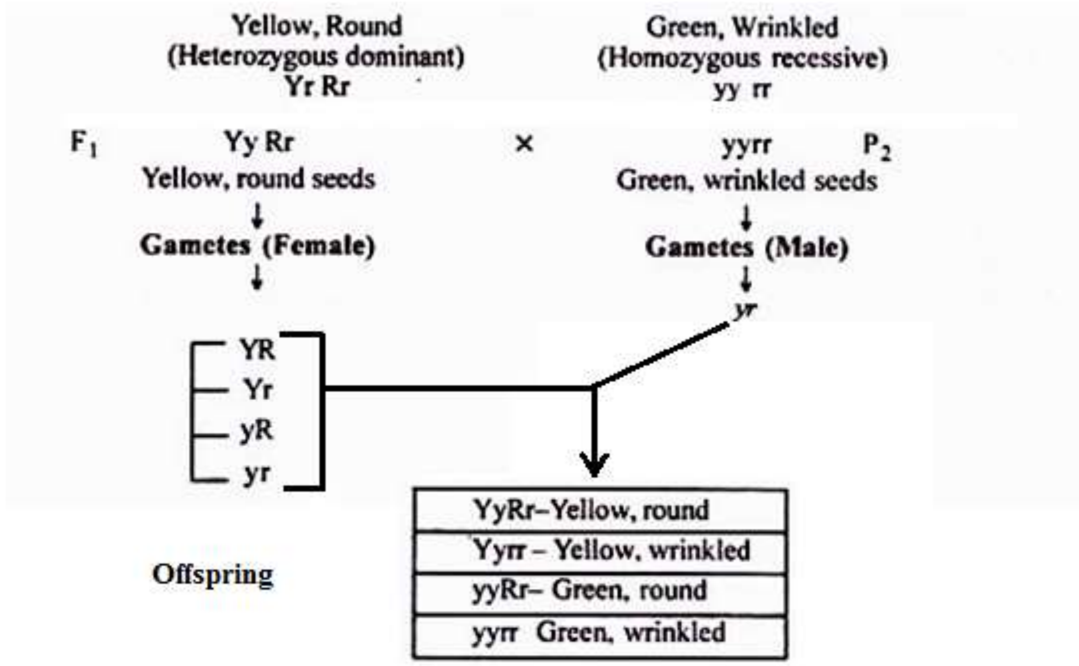
Mendel performed ample dihybrid crosses and reciprocal crosses with different combinations. Every time he got the same pattern of the result.

The uniform expression was both dominant in F1 generation. In F2 generation always he got both dominant in large number.

Dihybrid test cross

- **DEFINITION:** A test cross is the cross of any individual to a homozygous recessive parent; used to determine if the individual is homozygous dominant or heterozygous.
  - If all offspring are of one variety then it means the parent is homozygous dominant.
  - If we obtain offspring of four varieties (in the ratio 1:1:1:1) then it means the parent is heterozygous.
- **Dihybrid test cross** and it gives a **phenotypic ratio** of **1:1:1:1**.
- Below is an example of a dihybrid test cross involving seed colour and seed shape in peas.

1. **Offspring of four varieties in the ratio 1:1:1:1 means parent is heterozygous:**



1/4 Yellow, round; 1/4 Yellow, wrinkled; 1/4 Green, round; 1/4 Green, wrinkled

Phenotypic ratio 1:1:1:1

Genotypic ratio 1 Yy Rr : 1 Yy rr : 1 yy Rr : 1 yy rr

2. All offspring of one variety means the parent is homozygous dominant.

✍**Task:** Complete the genetic diagram for this test cross.

Mendel's Third Law of Inheritance – Law (Principle) of Independent Assortment:

- For two characteristics, the genes are inherited independently of one another.
- Also called the "Inheritance Law"
- For example: If you had the genotype **RrYy** you would make four kinds of gametes: they would contain the combinations of either **RY**, **Ry**, **rY** or **ry**.
- In other words, when a dihybrid (or polyhybrid) forms gametes, assortment (distribution) of alleles or different traits is independent of their original combinations in the parents. This law can be explained by help of dihybrid cross and dihybrid ratio.

Importance of Mendel's Laws:

- The concept of dominant and recessive factors is very important. This character is shown by many hereditary traits.
- It gives an idea of new combinations of traits which are very useful in developing a desirable trait in a progeny.
- This information is particularly used in the field of plant and animal breeding. Thus new types of plants and animals can be produced by hybridization.

SUMMARY: Mendel's Dihybrid cross

P1 phenotypes: Round yellow X Wrinkled green
P1 genotypes (2n): RRYY X rryy
Meiosis:
Gametes (n): All RY All ry
Random Fertilisation:
F1 genotypes (2n): All RrYy
F1 phenotypes All heterozygous round yellow seeds

*** F1 offspring were selfed**
F1 phenotypes: Round yellow seeds X Round yellow seeds
F1 genotypes (2n): RrYy RrYy

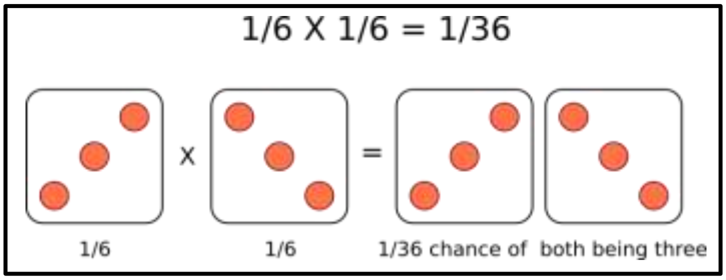
Gametes (n) and Random fertilisation shown by Punnett square:

	RY	rY	Ry	ry
RY	RRYY Round yellow	RrYY Round yellow	RRYy Round yellow	RrYy Round yellow
rY	RrYY Round yellow	rrYY Wrinkled yellow	RrYy Round yellow	rrYy Wrinkled yellow
Ry	RRYy Round yellow	RrYy Round yellow	RRyy Round green	Rryy Round green
ry	RrYy Round yellow	rrYy Wrinkled yellow	Rryy Round green	rryy Wrinkled green

F2 genotypes (2n) (as listed in each square)
F2 genotypic ratio 1RRYY: 2RrYY: 2RRYy: 4RrYy: 1rrYy: 2rrYy: 1RRyy: 2Rryy: 1rryy)
F2 phenotypes **9 round yellow: 3 round green: 3 wrinkled yellow: 1 wrinkled green**

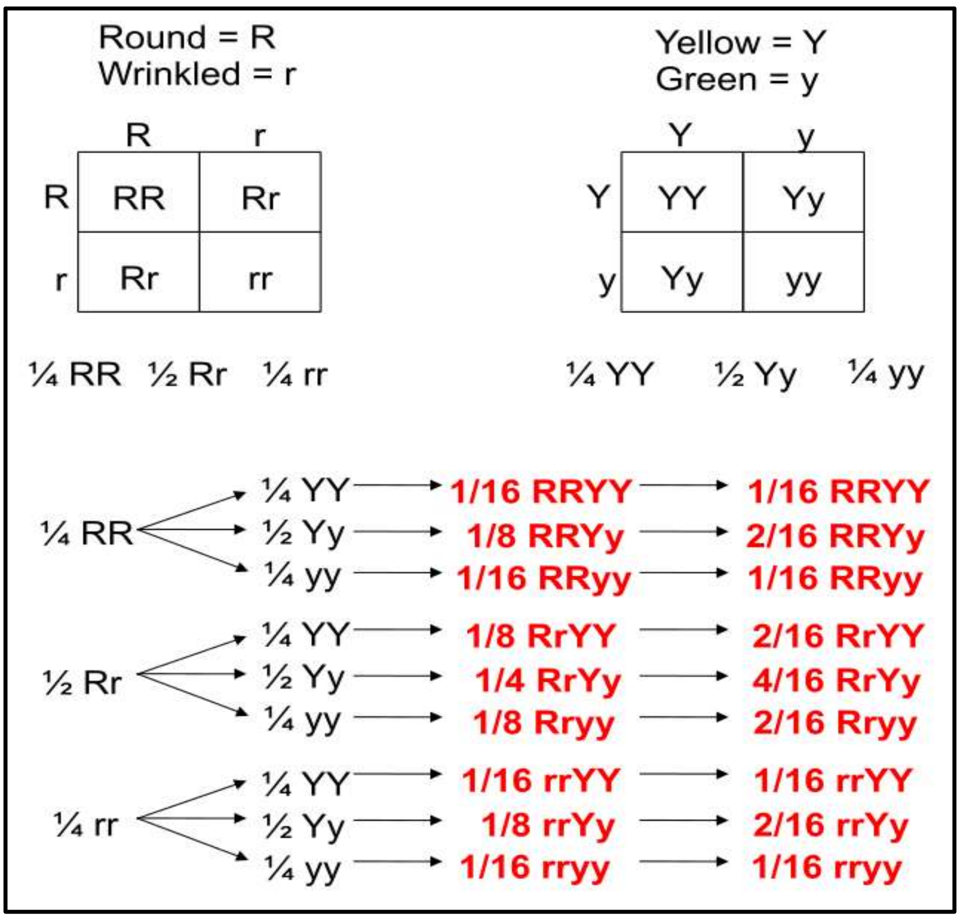
Calculating Probability from Punnett Squares

Punnett Squares are convenient for predicting the outcome of monohybrid or dihybrid crosses. The expectation of two heterozygous parents is 3:1 in a single trait cross or 9:3:3:1 in a two-trait cross. Performing a three or four trait cross becomes very messy. In these instances, it is better to follow the rules of probability. **Probability** is the chance that an event will occur expressed as a fraction or percentage. In the case of a monohybrid cross, 3:1 ratio means that there is a ¾ (i.e. 0.75 or 75%) chance of the dominant phenotype with a ¼ (i.e. 0.25 or 25%) chance of a recessive phenotype.



A single die has a 1 in 6 chance of being a specific value. In this case, there is a 1/6 probability of rolling a 3. It is understood that rolling a second die simultaneously is not influenced by the first and is therefore independent. This second die also has a 1/6 chance of being a 3.

We can understand these rules of probability by applying them to the dihybrid cross and realizing we come to the same outcome as the 2 monohybrid Punnett Squares as with the single dihybrid Punnett Square.



This forked line method of calculating probability of offspring with various genotypes and phenotypes can be scaled and applied to more characteristics.

Activity: To demonstrate Dihybrid inheritance of Maize/Corn (*Zea mays*) Colour and texture in an F_2 Population

KEY: A. Purple and smooth. B. Purple and shrunken
C. Yellow and smooth. D. Yellow and shrunken.

1. Retrieve a dihybrid F_2 maize/corn cob
2. Count a total of 200 kernels

a) Tally the number of Yellow Kernels that are smooth in texture.
b) Tally the number of Yellow Kernels (honey-colored) that are wrinkled in texture.
c) Tally the number of Purple Kernels that are smooth in texture.
d) Tally the number of Purple Kernels that are wrinkled in texture.
e) Ignore any speckled kernels that may have yellow and purple within them.

3. Compare numbers with the class as a whole.
4. Each kernel constitutes an individual organism (a seed that can give rise to a whole new plant). From the numbers:

a) What is the dominant colour? (yellow or purple)?
b) Is there a dominant texture (smooth or wrinkled)?
c) What is the dominant texture, if there is?
d) Is there a color that always pairs with a texture or do these characteristics *assort independently*?
e) What is the phenotypic ratio of purple and smooth: purple and wrinkled: yellow and smooth: yellow and wrinkled.
f) What name is given to this ratio?
g) Create a Punnett square to illustrate the expected number of each color/texture combination in a simple dominant: recessive paradigm.

TYPICAL RESULTS

Kernel type	Tally	Number
Purple and smooth	卌 II	112
Purple and wrinkled	卌 卌 卌 卌 卌 卌 卌 II	37
Yellow and smooth	卌 卌 卌 卌 卌 卌 卌 III	38
Yellow and wrinkled	卌 卌 III	13

a) What is the dominant colour? (Yellow or purple)? Purple.
b) Is there a dominant texture (smooth or wrinkled)? Yes.
c) What is the dominant texture, if there is? Smooth.

(**NB:** The biggest tally or number of offspring corresponds to the dominant traits. Therefore the dominant traits are Purple and smooth. The smallest tally or number of offspring corresponds to the recessive traits. Therefore the recessive traits are Yellow and wrinkled).

d) Is there a color that always pairs with a texture or do these characteristics *assort independently*? They assort independently.
e) What is the phenotypic ratio of purple and smooth: purple and wrinkled: yellow and smooth: yellow and wrinkled.

112:37:38:13.
Divide by smallest number.
112/13: 37/13: 38/13: 13/13 = 8.62: 2.85: 2.92: 1. This is approximately 9:3:3:1.

f) What name is given to this ratio? Dihybrid ratio.
g) Create a Punnett square to illustrate the expected number of each color/texture combination in a simple dominant: recessive paradigm.

P =Purple (dominant)
p = Yellow (Recessive)
S = Smooth (Dominant)
s =wrinkled (Recessive)

Parental (F1) phenotype: Purple and smooth X Purple and smooth
Parental (F1) genotypes (2n): PpSs X PpSs

Gametes (n) and random fertilisation as shown by Punnett square:

	PS	Ps	pS	ps
PS	PPSS	PPSs	PpSS	PpSs
Ps	PPSs	PPss	PpSs	Ppss
pS	PpSS	PpSs	ppSS	ppSs
ps	PpSs	Ppss	ppSs	ppss

F2 genotypes: (As listed in each square)
F2 genotypic ratio: 1PPSS:2PpSS:2PPSs:4PpSs:1PPss:2Ppss:1ppSS:2ppSs:1ppss
F2 phenotypes:
9 purple smooth:3 purple wrinkled:3 yellow smooth:1 yellow wrinkled

Mendel and Meiosis

Gregor Mendel demonstrated that the inheritance of traits (i.e. genes) follow particular laws:

- **Law of Segregation:** Each hereditary characteristic is controlled by two alleles which separate into different gametes
- **Law of Independent Assortment:** The separation of alleles for one gene is independent to allele separation for another gene
- **Principle of Dominance:** In pairs of alleles that are different, one allele will mask the effect of the other allele

These 'laws' are now known to be due to key events that occur during meiotic division:

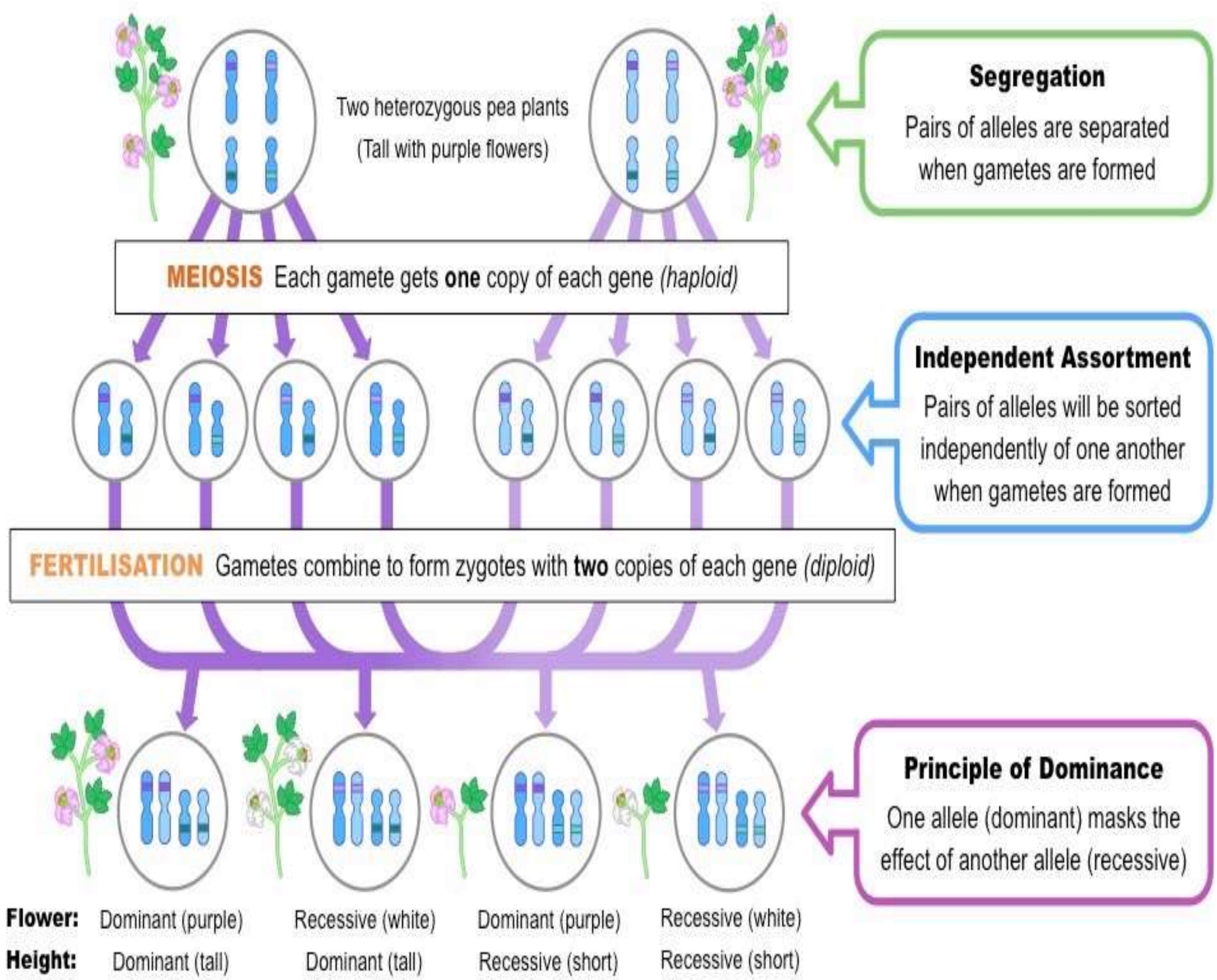
- The *law of segregation* describes how homologous chromosomes (and hence allele pairs) are separated in meiosis I
- The *law of independent assortment* describes how homologous pairs align randomly (as bivalents) during metaphase I

Deviations from Mendel's laws:

Through the elucidation of the process of meiosis, we now know that there are certain exceptions to Mendel's laws:

- Genes that are on the same chromosome (*linked genes*) will **not** undergo independent assortment (unless recombination occurs). This is deviation from the Law of independent assortment.
- Not all genes display a dominance hierarchy – certain traits may display codominance or incomplete dominance. This is also deviation from the law of dominance.

The Role of Meiosis in Mendelian Inheritance



DEVIATIONS FROM MENDELIAN GENETICS.

- Deviations from Mendelian genetics results in what is known as Non-Mendelian genetics.
- Incomplete dominance and Codominance are examples of deviations from Mendel's law of dominance.
- **Do all of these exceptions invalidate Mendel's laws?**

1. No. Mendel's laws explain the basic situation, and all of these exceptions are best understood in light of the mechanisms that Mendel described. Scientists generally try to understand simple cases before moving on to the more baffling ones, and often (as here) understanding the simpler cases helps form the basis for understanding the more complicated ones.
2. However…it is important to know about these "exceptions" and apparent exceptions, because most genetic inheritance has some aspect of at least one of these "exceptions" in it.

INCOMPLETE DOMINANCE – Blending of parental traits

- In many ways Gregor Mendel was quite lucky in discovering his genetic laws. He happened to use pea plants, which happened to have a number of easily observable traits that were determined by just two alleles. And for the traits he studied in his peas, one allele happened to be dominant for the trait & the other was a recessive form. Things aren't always so clear-cut & "simple" in the world of genetics, but luckily for Mendel he happened to work with an organism whose genetic make-up was fairly clear-cut & simple.
- If Mendel were given a mommy black mouse & a daddy white mouse & asked what their offspring would look like, he would've said that a certain percent would be black & the others would be white. He would never have even considered that a white mouse & a black mouse could produce a *GREY* mouse! For Mendel, the phenotype of the offspring from parents with different phenotypes always resembled the phenotype of at least one of the parents. In other words, Mendel was unaware of the phenomenon of INCOMPLETE DOMINANCE.
- In incomplete dominance, F_1 generation has a phenotype that does not resemble either of the two parents, but is a mixture/blend of the two.
- With **incomplete dominance**, a cross between organisms with two different phenotypes produces offspring with a **third phenotype** that is a **blending** of the parental traits.

- Flower colour in snapdragon (dog flower), *Antirrhinum majus*, is an example of Incomplete Dominance and is like this:

RED Flower x WHITE Flower ---> PINK Flower

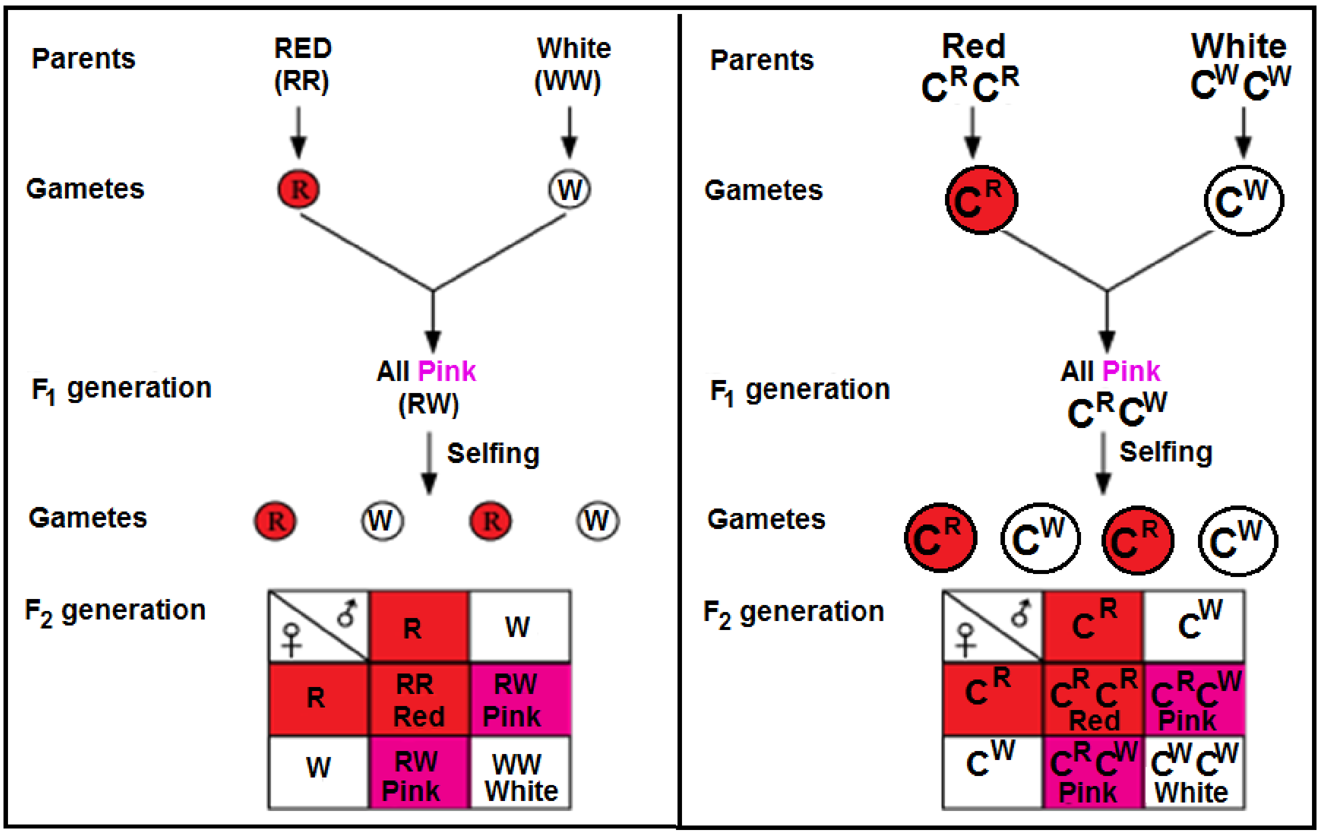
where:

- RR – Red flowers
- WW – White flowers
- RW – Pink flowers

- It's like mixing paints, red + white will make pink. Red doesn't totally block (dominate) white, instead there is *incomplete* dominance, and we end up with something in-between (intermediate colour) i.e. pink.
- The basis for the intermediate color (Pink) in the heterozygote is simply that the pigment produced by the red allele (anthocyanin) is diluted in the heterozygote and therefore appears pink because of the white background of the flower petals.

- Snapdragon flower/petal colour is controlled by gene C with two alleles, written with superscripts as C^R and C^W or sometimes simply as R and W.
- C^R is the allele for red petals/flowers and C^W is the allele for white petals.
- When a cross is made between red-flowered snapdragon (C^RC^R) and white-flowered snapdragon (C^WC^W) varieties, F1 (C^RC^W) offspring produced is all pink flowered. When these F1 pink-flowered are self-pollinated or crossed among themselves to raise F2 generation, they produce red (C^RC^R), pink (C^RC^W) and white (C^WC^W) flowers in the 1:2:1 ratio respectively. This phenotypic ratio is identical with genotypic ratio because heterozygotes are phenotypically intermediate between two homozygous types.
  – Phenotypic Ratio – 1 Red : 2 Pink : 1 White.
  – Genotypic Ratio – 1 C^RC^R : 2 C^RC^W : 1C^WC^W. (Or simply 1 RR : 2 RW : 1 WW).

- This is represented in the cross diagrams that follow.
- There are two ways of representing the cross. The diagram on the right uses alleles written in superscript form (use any one of these in the exam!!):



- Phenotypic Ratio – 1:2:1 that denotes Red: Pink: White
- Genotypic Ratio – 1:2:1 that denotes RR: RW: WW respectively.

These pink flowers of a heterozygote snapdragon result from incomplete dominance.

(credit: "storebukkebruse"/Flickr)

OTHER EXAMPLES OF INCOMPLETE DOMINANCE

Flower/petal colour in Four o'clock plant (*Mirabilis jalapa*)

In Four o'clock plant (*Mirabilis jalapa*) just as in snapdragon or dog flower (*Antirrhinum majus*), there are three types of flower colours i.e., red, Pink and white

- When a cross is made between red-flowered Four o'clock plant (C^RC^R) and white-flowered Four o'clock plant (C^WC^W) varieties, F1 (C^RC^W) offspring produced is all pink flowered. When these F1 pink-flowered are self-pollinated or crossed among themselves to raise F2 generation, they produce red (C^RC^R), pink (C^RC^W) and white (C^WC^W) flowers in the 1:2:1 ratio respectively. This phenotypic ratio is identical with genotypic ratio because heterozygotes are phenotypically intermediate between two homozygous types.
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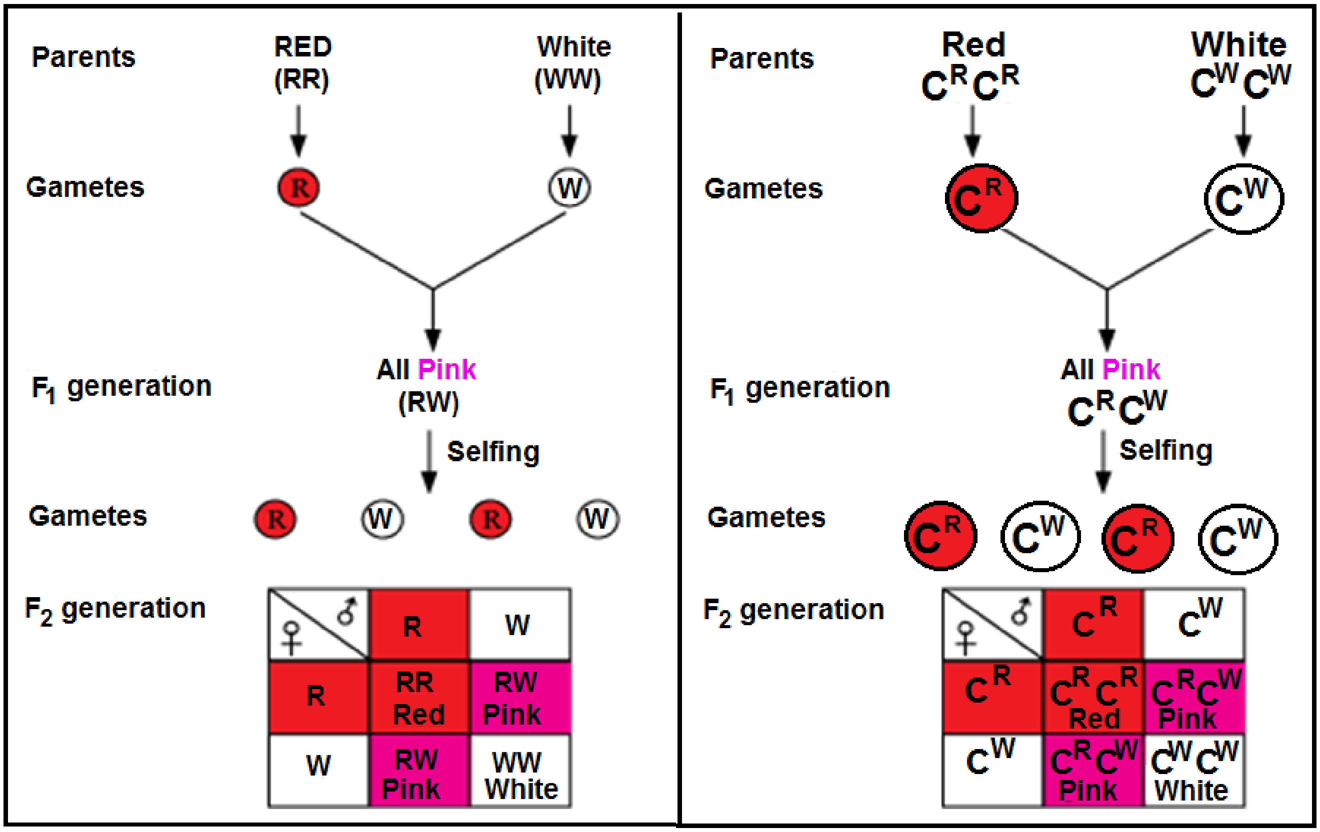


- Phenotypic Ratio – 1:2:1 that denotes Red: Pink: White
- Genotypic Ratio – 1:2:1 that denotes RR: RW: WW respectively.

Colour of feathers in Andalusian fowl

The Andalusian fowl is found in three colours: Black, white and blue. Pure forms (homozygotes) are black (BB) and white (WW). If these two forms are crossed, F_1 individuals appear blue (BW) coloured. The blue hybrids on crossing with each other (BW x BW) give rise to black, blue and white in 1:2:1 ratio respectively.

✍ **TASK:** Present the information above about colour of feathers in Andalusian chickens using genetic diagrams.

Interpreting incomplete dominance questions

The trick is to *recognize* when you are dealing with a question involving incomplete dominance. There are two steps to this:

1) Notice that the offspring is showing a **3rd phenotype**. The parents each have one, and the offspring are different from the parents.

2) Notice that the trait in the offspring is a **blend (mixing)** of the parental traits.

Sample Questions

1. A cross between a blue blahblah bird & a white blahblah bird produces offspring that are silver. The color of blahblah birds is determined by just two alleles.

a) What are the genotypes of the parent blahblah birds in the original cross?

b) What is/are the genotype(s) of the silver offspring?

c) What would be the phenotypic ratios of offspring produced by two silver blahblah birds?

2. The color of fruit for plant "X" is determined by two alleles. When two plants with orange fruits are crossed the following phenotypic ratios are present in the offspring: 25% red fruit, 50% orange fruit, 25% yellow fruit. What are the genotypes of the parent orange-fruited plants?

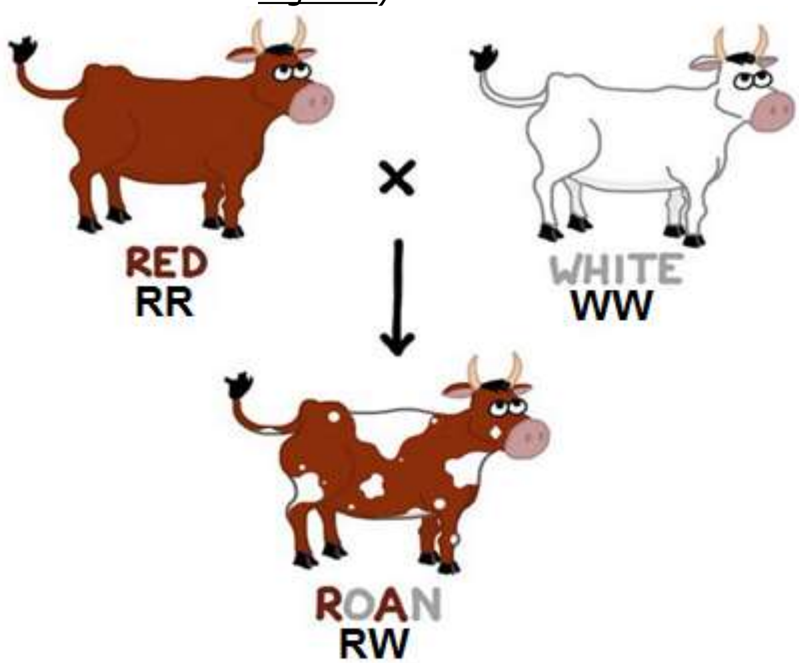
CO-DOMINANCE – Offspring resembling both parents (both of parental traits appear together in offspring)

- First let me point out that the meaning of the prefix "co-" is "together". Cooperate = work together. Coexist = exist together. Cohabitat = habitat together.
- With **codominance**, a cross between organisms with two different phenotypes produces offspring with a third phenotype in which both of the parental traits appear together.
- That is, with codominance, a cross between two different homozygotes produces heterozygotes in which both parental traits are expressed together.
- DEFINITION: Codominance is whereby two alleles fail to dominate each other in heterozygosis and their effects are expressed together in the offspring.
- In co-dominance, the F_1 progeny resembles both the parents.

Roan fur in short-horn cattle is an example of codominance and is like this:

red x white ---> red & white spotted

i.e. short-horn cattle fur can be red (RR = all red hairs), white (WW = all white hairs), or roan (RW = red & white hairs together).



- Roan bull crossed with a white cow/heifer produces all roan F1 offspring.

(ii) Red allele – R, white – ~~W~~; (A) Different symbols. [1]

(A) – C^R C^W

(b) Parental phenotypes: Red bull x white cow

Parental genotypes: RR x WW; (1 mark) for genotypes

Meiosis

Gametes: (R) (R) x (W) (W), (1 mark) for gametes

Random Fertilisation:

Offspring genotype: RW RW RW RW; , (1 mark) for genotypes

Offspring phenotype: roan roan roan roan;

all roan

- The roan F1 offspring selfed/crossed will produces red, roan and white offspring in the ration 1 red: 2 roan: 1 white.

(c) Parental phenotypes: roan x roan

Parental genotypes: RW x RW

Meiosis

Gametes: (R) (W) x (R) (W) (1 mark)

Offspring genotype: RR RW RW WW; (1 mark)

Offspring phenotype: Red roan roan white; (1 mark)

Phenotype ratio: 1 : 2 : 1; (1 mark)

[4]

Try these questions

1. Predict the phenotypic ratios of offspring when a homozygous white cow is crossed with a roan bull.
2. What should the genotypes & phenotypes for parent cattle be if a farmer wanted only cattle with red fur?

Well, the only way to have red fur is to be homozygous red (RR). In order to get that genotype in all the offspring both parents must be "RR". A parent with one or more "W" alleles will cause the inheritence of roan fur in some offspring. Go ahead & work out all the punnett squares if you don't believe me.
Only RR x RR gives you 100% RR.
RR x RW would produce 50% roan, 50% red,
RW x RW produces 25% red, 50% roan & 25% white,
WW x RW would produce 50% roan, 50% white,
& WW x RR would produce 100% roan (RW).

Other examples of codominance

Human blood type AB

- Another example of codominance is human blood type AB, in which two types of protein ("A" & "B") appear together on the surface of blood cells.
- ABO blood groups are controlled by gene I. Gene I has three alleles, I^A, I^B and i^O. A person possesses any two of the three alleles.
- The i^O allele is sometimes simply written as i.
- I^A and I^B dominate over i^O. But with each other, **I^A and I^B are co-dominant**.
- I^A and I^B contain A and B types of sugar, while i does not contain any sugar.

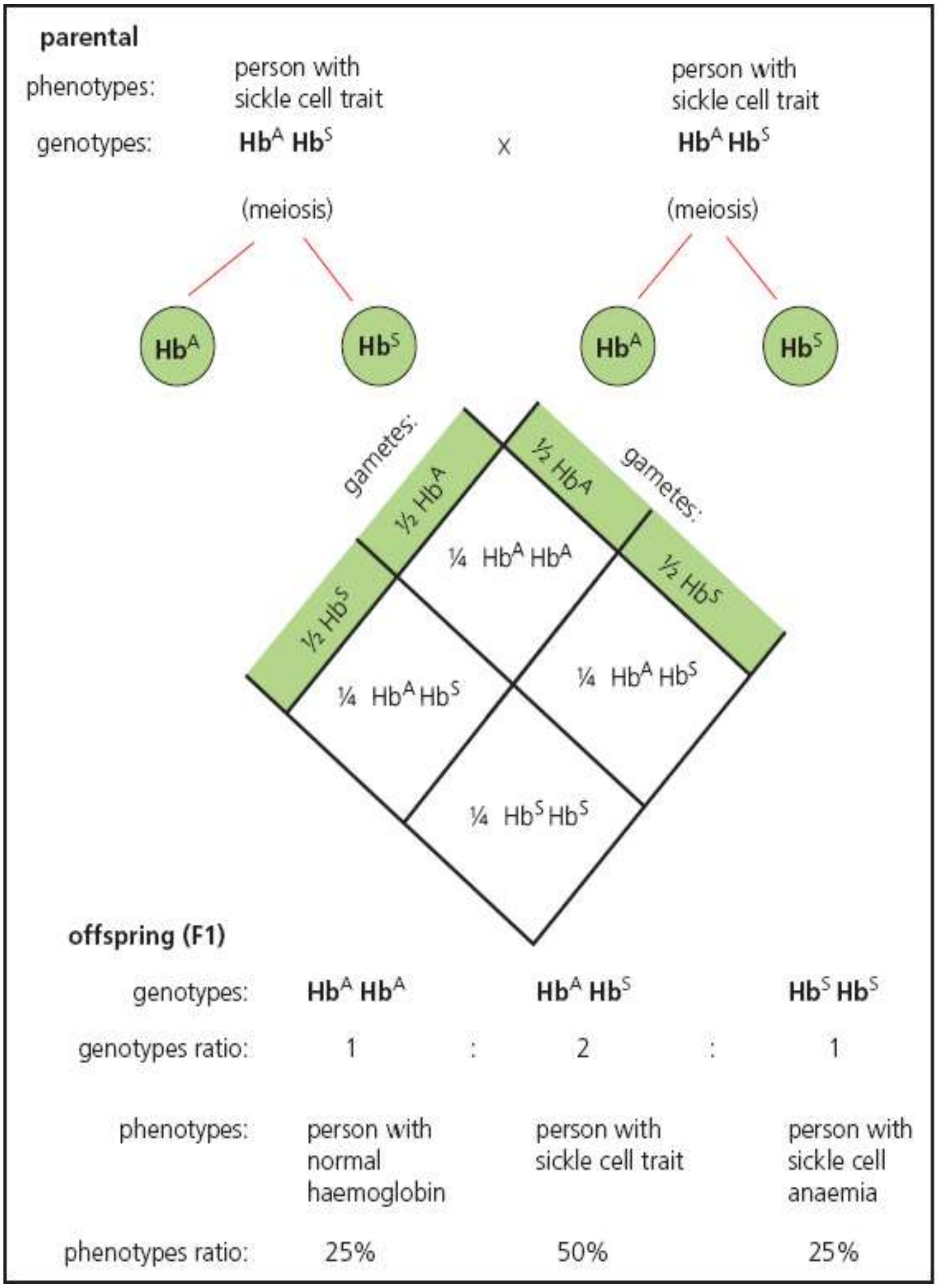
Allele from Parent 1	Allele from Parent 2	Genotype of offspring	Blood type of offspring
I^A	I^A	$I^A I^A$	A
I^A	I^B	$I^A I^B$	AB
I^A	i^o	$I^A i^o$	A
I^B	I^A	$I^A I^B$	AB
I^B	I^B	$I^B I^B$	B
I^B	i^o	$I^B i^o$	B
i^o	i^o	$i^o i^o$	O

- **Multiple alleles** occur when more than two alleles control a character, as in human blood groups.
  - Multiple alleles are used in population studies.

Sickle cell haemoglobin

- Another example of codominance is sickle cell haemoglobin in humans. The gene for haemoglobin Hb has two codominant alleles: Hb^A (the normal gene) and Hb^S (the mutated gene). There are three phenotypes:
- Hb^AHb^A Normal. All haemoglobin is normal, with normal red blood cells.
- Hb^AHb^S Sickle cell trait. 50% of the haemoglobin in every red blood cell is normal, and 50% is abnormal. The red blood cells are slightly distorted, but can carry oxygen, so this condition is viable. However these red blood cells cannot support the malaria parasite, so this phenotype confers immunity to malaria.

- $Hb^S Hb^S$ Sickle cell anaemia. All haemoglobin is abnormal, and molecules stick together to form chains, distorting the red blood cells into sickle shapes. These sickle red blood cells are destroyed by the spleen, so this phenotype is fatal.
- The genetic diagram that follows shows the inheritance of sickle cell haemoglobin – an example of codominance.



Task:

(a) (i) Describe the mutation in haemoglobin that causes red blood cells to be sickle-shaped.
(ii) What are the symptoms of sickle cell anaemia?

(a) Using a full genetic diagram show how sickle cell haemoglobin is inherited.

(b) (i) Why is the $Hb^A Hb^S$ genotype referred to as codominance?

(iii)Why does this genotype confer immunity to malaria?
(iv) Why is the Hb^SHb^S genotype fatal?

Try this question

A cross between a black cat & a tan cat produces a tabby pattern (black & tan fur together).

a) What pattern of inheritence does this illustrate?
Codominance, two phenotypes <u>together</u> at the same time.

b) What percent of kittens would have tan fur if a tabby cat is crossed with a black cat?

Tabby cats are the hybrids (because they have both colors) & a black cat must be homozygous black.
So the cross for this problem is BB (black) x BT (tabby).

The p-square is at the right.

The results show that 50% of the offspring will be BB (black) & 50% will be tabby (BT). So to answer the question, 0% of the kittens will be tan.

	B	T
B	BB	BT
B	BB	BT

THE CHI-SQUARED (χ^2) TEST

❖ **By the end of this section you should be able to:**

1. use the chi-squared test to test the significance of differences between observed and expected results.

❖ One of the most important statistical tests you can carry out in genetics is the **chi-squared (χ^2) test** (also known as **Pearson's chi-squared test** because it was developed by **Karl Pearson**).

❖ **Why would you need to carry out a χ^2-test?**

- Let's go all the way back to our **yellow/green**, **round/wrinkled peas**:
  - From our **theoretical crosses**, we figured out that when you **cross two heterozygous parents (*RrYy*)**, you would **EXPECT** to see a **phenotypic ratio** of **9** round yellow peas : **3** round, green peas : **3** wrinkled yellow peas : **1** wrinkled green pea in the offspring.
- **But what if we actually did this cross for real?** Would the **numbers** we were **EXPECTING** to see **be the same as the numbers** we'd actually **OBSERVE** in real life? **And how would you PROVE that what you did see wasn't down to random chance? That's where your χ^2–test comes in!**
  - ➢ So **chi-squared (χ^2) test** is a **statistical test** which allows us **to compare** our **observed** (actual) results with what we would **expect** if the **phenotypic ratio** held

true (e.g. **Mendelian dihybrid 9:3:3:1 ratio or some other phenotypic ratio such as the 1:1:1:1 testcross ratio etc**).

- It allows us **to objectively say** whether our **observed** results are as we **expected**.
- Chi-squared is a measure of the combined error of our results from the expected.
- It will always be conducted from a dihybrid cross at A-level.

❖ **Let's say we did our cross, counted the numbers of peas and we actually got:**

Yellow Round Peas	Green Round Peas	Yellow Wrinkled Peas	Green Wrinkled Peas
219	81	69	31

- **Is that a 9:3:3:1 ratio? Possibly, but let's prove it.**

- Because this is a test, you have to have a **hypothesis** that can be **tested** and **proved** either **right** or **wrong**: for the χ^2–test, this is known as the **null hypothesis (H_0)**.
- The null hypothesis basically states that there is no difference between the results you observed and the ones you were expecting, in this case a 9:3:3:1 ratio.

So first, we must write up a Null Hypothesis (H_0).

- **A null hypothesis always states that**

'There is NO significant difference between the observed and expected result. Any variation is totally due to chance events'

❖ **But how do we do that mathematically?**

Here's how: **below is the formula for Chi-squared**

$$\text{Chi-squared, } \chi^2 = \sum \frac{(O-E)^2}{E}$$

where O = Observed

E = Expected

Σ = sum of

❖ Little bit scary at first sight, isn't it? Let's break it down into its component parts so it doesn't look so bad:

- **First**, we need to **calculate the number of each phenotype** we would **expect to see** if we did get a perfect 9:3:3:1 ratio.

❖ You do this by finding the total number of your actual peas and multiplying it by the expected ratio.

Total number of actual peas = 219 + 81+ 69 + 31 = **400**
Total ratio = 9 + 3 + 3 + 1 = 16

- 9/16 Yellow round = 9/16 x 400 = 225
- 3/16 Green round = 3/16 x 400 = 75

- 3/16 Yellow wrinkled = 3/16 x 400 =75
- 1/16 Green wrinkled = 1/16 x 400 = 25

❖ **Now we complete the following table**:

Phenotype	Observed (O)	Expected (E)	O – E	$(O-E)^2$	$(O-E)^2 / E$
Round Yellow Peas	219	225			
Round Green Peas	81	75			
Wrinkled Yellow Peas	69	75			
Wrinkled Green Peas	31	25			

- Next, we need to work out the difference between the Observed and Expected (O – E), and then (to get rid of any inconvenient negative numbers), square that result.

Phenotype	Observed (O)	Expected (E)	O – E	$(O-E)^2$	$(O-E)^2 / E$
Round Yellow Peas	219	225	-6	36	
Round Green Peas	81	75	6	36	
Wrinkled Yellow Peas	69	75	-6	36	
Wrinkled Green Peas	31	25	6	36	
				$\chi^2 = \Sigma$	

- Now we can calculate the rest of the equation, by dividing the $(O-E)^2$ result by the Expected number (E) for that phenotype and adding them all together:

Phenotype	Observed (O)	Expected (E)	O – E	$(O-E)^2$	$(O-E)^2 / E$
Round Yellow Peas	219	225	-6	36	0.16
Round Green Peas	81	75	6	36	0.48
Wrinkled Yellow Peas	69	75	-6	36	0.48
Wrinkled Green Peas	31	25	6	36	1.44
				$\chi^2 = \Sigma$	**2.56**

❖ So for our example, **the calculated chi-squared (χ^2) value = 2.56**.
- But it's just a number: **how does it prove or disprove our null hypothesis**, that there is no difference between the observed and expected results?

❖ Now we need to bring in another factor, something known as **the degrees of freedom (df).** These take into account the number of different classes represented by the cross, in this case = 4 different phenotypes. The more classes you have, the greater the variation you are likely to see from any expected numbers.

- **Calculating the degrees of freedom (df) is straight forward.**
  - It is **the number of actual different phenotypic groups** for the cross **(n)** **minus one.** The formula looks like this; **df = n – 1**
  - Therefore in this case **df = 4 – 1 = 3**

- Nearly there! Now we have our **calculated χ^2 value (2.56)** and **our degrees of freedom (3)**, we then go and take a look at a χ^2 **probability table** (there's an example of one here (see below)).
- **We want to know if our result is statistically significant**, and for most genetics tests, **a significance (or confidence) level of 0.05** is used (which means that **95 times out of 100**, you would see exactly what you were **EXPECTING** to see, with variation occurring **5 times out of 100** purely **due to random chance**).

- **What is the critical value?** Critical value of $\chi 2$, where p=0.05, is the level at which we are 95% certain that the result is not due to chance.
- If you look at the example χ^2 **probability table** for a **critical value** at **0.05** (i.e. 5%) **significance level** (along the top) **with 3 degrees of freedom** (down the left hand side), **the χ^2 value = 7.81.**
  - **Step by step, how do we do it?**
  - **Select the correct row of critical values** to **compare** your **chi-squared value against**.
  - Each row indicates the **level of degrees of freedom**. In this case it is row number 3 as we have 3 degrees of freedom.
  - Then we **move along the critical values** and check that our **chi-squared value** is **higher** or **lower** than the **critical value at 0.05** (5%) **significance level**.

- If the **calculated chi-squared value** is **higher** than the critical value at the 0.05 significance level then we **reject the null hypothesis**,
  - **Stating that**
    'Observed results are significantly different from expected results'

- If the **calculated chi-squared value** is **lower** than the critical value at the 0.05 significance level then **we accept the null hypothesis**,

  - **Stating that**
    'Observed results are not significantly different from expected results. Any variation is due to chance'.

- **Back to our example:**
  - Because our calculated value (of 2.56) is **lower** (less) **than the critical value** (of 7.81) we **can accept our null hypothesis** and say with **95% confidence** that **there is no significant difference between the expected and observed results!**
  - If our χ^2 value had been greater than 7.81, then we would have to accept that there was a statistically significant difference between the numbers we observed and the numbers we were expecting, and that yellow/green and round/wrinkled were not behaving in the way we had predicted.

Probability (p-) value of 0.25 = 25% chance that the difference is due to chance

TABLE C: χ^2 CRITICAL VALUES

Probability (p-) value of 0.05 = 5% chance that the difference is due to chance

df	Tail probability *p* .25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001
1	1.32	1.64	2.07	2.71	3.84	5.02	5.41	6.63	7.88	9.14	10.83
2	2.77	3.22	3.79	4.61	5.99	7.38	7.82	9.21	10.60	11.98	13.82
3	4.11	4.64	5.32	6.25	7.81	9.35	9.84	11.34	12.84	14.32	16.27
4	5.39	5.99	6.74	7.78	9.49	11.14	11.67	13.28	14.86	16.42	18.47
5	6.63	7.29	8.12	9.24	11.07	12.83	13.39	15.09	16.75	18.39	20.51
6	7.84	8.56	9.45	10.64	12.59	14.45	15.03	16.81	18.55	20.25	22.46
7	9.04	9.80	10.75	12.02	14.07	16.01	16.62	18.48	20.28	22.04	24.32
8	10.22	11.03	12.03	13.36	15.51	17.53	18.17	20.09	21.95	23.77	26.12
9	11.39	12.24	13.29	14.68	16.92	19.02	19.68	21.67	23.59	25.46	27.88
10	12.55	13.44	14.53	15.99	18.31	20.48	21.16	23.21	25.19	27.11	29.59
11	13.70	14.63	15.77	17.28	19.68	21.92	22.62	24.72	26.76	28.73	31.26
12	14.85	15.81	16.99	18.55	21.03	23.34	24.05	26.22	28.30	30.32	32.91
13	15.98	16.98	18.20	19.81	22.36	24.74	25.47	27.69	29.82	31.88	34.53
14	17.12	18.15	19.41	21.06	23.68	26.12	26.87	29.14	31.32	33.43	36.12
15	18.25	19.31	20.60	22.31	25.00	27.49	28.26	30.58	32.80	34.95	37.70
16	19.37	20.47	21.79	23.54	26.30	28.85	29.63	32.00	34.27	36.46	39.25
17	20.49	21.61	22.98	24.77	27.59	30.19	31.00	33.41	35.72	37.95	40.79
18	21.60	22.76	24.16	25.99	28.87	31.53	32.35	34.81	37.16	39.42	42.31
19	22.72	23.90	25.33	27.20	30.14	32.85	33.69	36.19	38.58	40.88	43.82
20	23.83	25.04	26.50	28.41	31.41	34.17	35.02	37.57	40.00	42.34	45.31
21	24.93	26.17	27.66	29.62	32.67	35.48	36.34	38.93	41.40	43.78	46.80
22	26.04	27.30	28.82	30.81	33.92	36.78	37.66	40.29	42.80	45.20	48.27
23	27.14	28.43	29.98	32.01	35.17	38.08	38.97	41.64	44.18	46.62	49.73
24	28.24	29.55	31.13	33.20	36.42	39.36	40.27	42.98	45.56	48.03	51.18
25	29.34	30.68	32.28	34.38	37.65	40.65	41.57	44.31	46.93	49.44	52.62
26	30.43	31.79	33.43	35.56	38.89	41.92	42.86	45.64	48.29	50.83	54.05
27	31.53	32.91	34.57	36.74	40.11	43.19	44.14	46.96	49.64	52.22	55.48
28	32.62	34.03	35.71	37.92	41.34	44.46	45.42	48.28	50.99	53.59	56.89
29	33.71	35.14	36.85	39.09	42.56	45.72	46.69	49.59	52.34	54.97	58.30
30	34.80	36.25	37.99	40.26	43.77	46.98	47.96	50.89	53.67	56.33	59.70
40	45.62	47.27	49.24	51.81	55.76	59.34	60.44	63.69	66.77	69.70	73.40
50	56.33	58.16	60.35	63.17	67.50	71.42	72.61	76.15	79.49	82.66	86.66
60	66.98	68.97	71.34	74.40	79.08	83.30	84.58	88.38	91.95	95.34	99.61

Calculated χ^2 value LOWER than critical value. Accept Null (H_0) hypothesis.

Observed results are not significantly different from expected results. Any variation is due to chance.

Critical value

Calculated χ^2 value HIGHER than critical value. Reject Null (H_0) hypothesis.

Observed results are significantly different from expected results.

Summary

Basically there are 4 or sometimes 5 stages (depending on type of question) followed when solving chi-squared statistical problems in genetics. The stages are:

1. Formulate the null (H_0) hypothesis.
2. Construct the table.
3. Statistical parameters.
4. Interpreting the result.
5. Conclusion.

See the worked examples that follow.

WORKED EXAMPLES

Worked example 1

A true heterozygous pea plant for round yellow seeds was crossed with another true heterozygous pea plant for round yellow seeds.

R – allele for round seeds

r – allele for wrinkled seeds

Y – allele for yellow seeds

y – allele for green seeds

RrYy	x	RrYy
Round yellow		Round yellow

The observed F1 results of pea plants were as follows.

- 9/16 Yellow round = 169
- 3/16 Green round = 54
- 3/16 Yellow wrinkled = 51
- 1/16 Green wrinkled = 14

Use the chi-squared (χ^2) test to test the significance of the difference between observed and expected results.

Solution 1

1. Formulate the null (H_0) hypothesis:

There is NO significant difference between the observed result and expected result. Any variation is totally due to chance.

2. Construct the table:

The total number of pea plants = 169 + 54 + 51 + 14 = 288
The expected number (E) in each case:

9/16 Yellow round = 9/16 x 288 = 162
3/16 Green round = 3/16 x 288 = 54
3/16 Yellow wrinkled = 3/16 x 288 = 54
1/16 Green wrinkled = 1/16 x 288 = 18

Enter the Expected numbers in the table

$$\text{Chi-squared, } \chi^2 = \sum \frac{(O-E)^2}{E}$$

where O = Observed
E = Expected
Σ = sum of

The difference between the observed (O) and expected (E) values is calculated (O-E) and squared $[(O-E)^2]$. This value is divided by the expected number (E) and the sum of these values gives the value χ^2.

Chi-squared test

	Observed (O)	Expected (E)	O - E	$(O-E)^2$	$\frac{(O-E)^2}{E}$
Yellow round	169	162	7	49	0.302
Green round	54	54	0	0	0.000
Yellow wrinkled	51	54	-3	9	0.167
Green wrinkled	14	18	-4	16	0.889

$$\sum = \boxed{1.358}$$

Therefore the Chi-squared (x^2) value is **1.358**

3. Statistical parameters:

Number of classes, n = 4
Degrees of freedom, df = n – 1
= 4-1
= 3

The level of significance or probability level (p) is 0.05, corresponding with 95% confidence limits.
Using chi-squared tables, for df = 3 and p = 0.05, the critical value for χ^2 = 7.81.

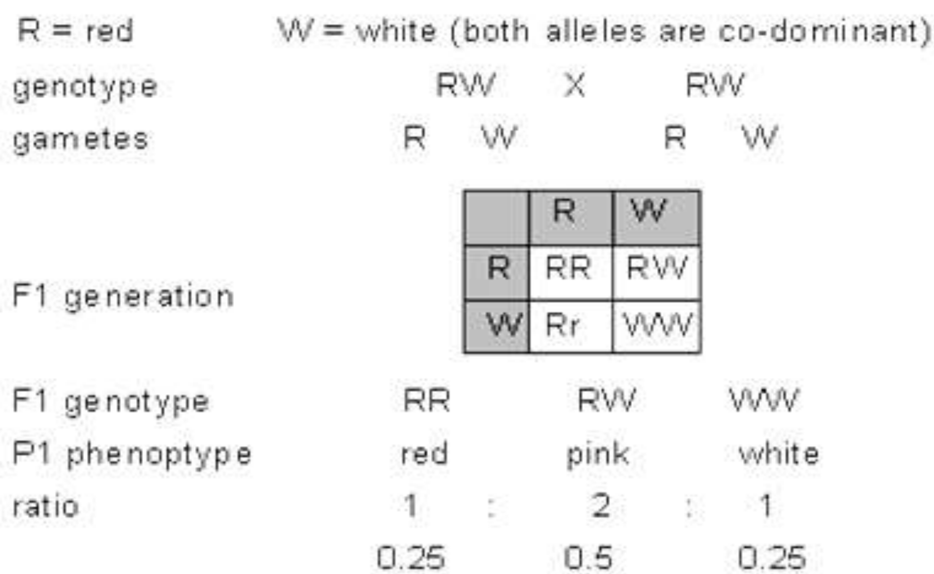
4. Interpreting the result:

The calculated value of **1.358** is **lower** (less) than **7.81**. Therefore **the null hypothesis** is accepted.

The observed results are not significantly different from expected results. Any variation is due to chance.

Worked example 2

Dianthus (campion) has flowers of three different colours, red, pink and white. Two pink flowered plants were crossed and the collected seeds grown to the flowering stage. The observed results of the cross where as follows: 34 red flower plants, 84 pink flower plants and 42 white flower plants. The cross is shown below. Use the chi-squared (χ^2) test to test the significance of the difference between observed and expected results.



1. Formulate the null (H_0) hypothesis:

There is NO significant difference between the observed result and expected result. Any variation is totally due to chance.

2. Construct the table:

Total number of observed plants = 34 + 84 + 42 = 160

The expected number in each case:

0.25 x 160 = 40
0.5 x 160 = 80
0.25 x 160 = 40

Phenotype	Observed (O)	Expected (E)	O – E	$(O-E)^2$	$\frac{(O-E)^2}{E}$
Red flowers	34	40	-6	36	0.9
Pink flowers	84	80	4	16	0.2
White flowers	42	40	2	4	0.1
				$\chi^2 = \Sigma$	**1.2**

3. Statistical parameters:
There are 3 classes (red, pink and white)

Degrees of freedom, df = 3 – 1 =2

4. Interpreting the result:

The nearest figures to 1.2 comes after 0.25 (or 25%). A χ^2 value of 1.2 shows that it is at least 25% probable that the result is by chance alone.

From the χ^2 **probability table** the critical value is **5.99** at 0.05 (or 5%) significance level.

The calculated value of **1.2** is **lower** (less) than **7.81**. Therefore **the null hypothesis** is accepted.

The observed results are not significantly different from expected results. Any variation is due to chance.

Worked example 3

The example given here is testing a dihybrid backcross in maize, in which the parents are tall, green with genotype TtGg, and dwarf albino, with genotype ttgg

Phenotype	Number of individuals	Possible ratio
tall green	31	1
tall albino	24	1
dwarf green	29	1
dwarf albino	16	1

Worked examples of statistical analysis
The Chi-squared test is suitable because there is a large sample and there are predicted values based on a null hypothesis, proposed based on Mendelian genetics.

1. Formulate the null hypothesis:
There is NO significant difference between the observed and expected phenotypic ratio which is 1 tall green : 1 tall albino : 1 dwarf green : 1 dwarf albino.
Any variation is totally due to chance.

2. Construct the table:

Total number of observed plants = 31 +24 + 29 +16 = 100
Total ratio 1+1+1+1 =4
The expected number in each case:

Tall green = 1/4 x 100 = 25
Tall albino = 1/4 x 100 = 25
Dwarf green = 1/4 x 100 = 25
Dwarf albino = 1/4 x 100 = 25

Phenotype	Observed	Expected	(O – E)	$(O - E)^2$	$\frac{(O-E)^2}{E}$
tall green	31	25	6	36	1.44
tall albino	24	25	-1	1	0.04
dwarf green	29	25	4	16	0.64
dwarf albono	16	25	-9	81	3.24
				$\chi^2 = \Sigma\frac{(O-E)^2}{E}$	5.36

Calculated χ^2 value = **5.36**

3. Statistical parameters:

(i) The number of degrees of freedom (df) is one less than the number of phenotypes, so df = 4 – 1 = 3.
(ii) The level of significance or probability level (p) is 0.05, corresponding with 95% confidence limits.
(iii) Using chi-squared tables, for df = 3 and p = 0.05, the critical value for χ^2 = 7.81.

4. Interpreting the result:

the calculated χ^2 value (5.36) is less than the critical value (7.81).

∴ the null hypothesis is accepted at the 0.05 level of significance

∴ the ratio is 1:1:1:1 and any deviation from the ideal ratio is due to chance.

5. Conclusion:

The phenotype of the sample is determined by two alleles each of two unlinked genes, where "tall" is dominant over "dwarf" and "green" is dominant over "albino".
The sample is an F2 generation, produced by crossing two heterozygous plants . Inheritance is Mendelian.

Applying the chi-squared test to Dihybrid inheritance of Maize/Corn (*Zea mays*) Colour and texture in an F_2 Population

Mendelian genetics explains how different ratios of phenotypes can be produced when the genetic composition of the parents is known. Such ratios can be observed if a large enough sample of progeny is counted. The numbers counted may be subjected to statistical testing,

using the χ^2 test, which indicates whether or not the observed ratio is statistically equivalent to the theoretical Mendelian ratio.
In maize, *Zea mays, a* single F_2 cob is covered in very many kernels, each of which contains a seed which is the result of a single fertilisation.
The kernels show a number of characteristics, for example texture and colour, as is shown in the picture below.

The phenotypes shown by the cob were identified and the kernels having each phenotype were counted. The results were entered in the table below.

Kernel type	Tally	Number
Purple and smooth	𝍸 II	112
Purple and wrinkled	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 II	37
Yellow and smooth	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 III	38
Yellow and wrinkled	𝍸 𝍸 III	13

a) Identify the theoretical Mendelian ratio that is closest to the counts that you have made.
b) Use the $\chi 2$ test and probability table to test if the observed ratio is statistically equivalent to the theoretical Mendelian ratio.
c) Deduce the genotype of the parent plants.

Solution

Null hypothesis:
There is NO significant difference between the observed and expected (theoretical) Mendelian phenotypic ratio which is 9 purple and smooth: 3 purple and wrinkled: 3 yellow and smooth: 1 yellow and wrinkled. Any variation is totally due to chance.
Table:
Total number of observed plants = 112 +37 + 38 +13 = 200
Total ratio 9+3+3+1 =16

The expected number in each case:

Purple and smooth = 9/16 x 200 = 112.5
Purple and wrinkled = 3/16 x 200 =37.5
Yellow and smooth = 3/16 x 200 =37.5
Yellow and wrinkled = 1/16 x 200 = 12.5

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Phenotype	Observed number (O)	Expected number (E)	(O – E)	$(O - E)^2$	$\frac{(O-E)^2}{E}$
Purple and smooth	112	112.5	-0.5	0.25	0.00222
Purple and wrinkled	37	37.5	-0.5	0.25	0.00667
Yellow and smooth	38	37.5	0.5	0.25	0.00667
Yellow and wrinkled	13	12.5	0.5	0.25	0.02000
				$\chi^2 = \Sigma$	**0.0356**

Calculated χ^2 value = **0.0356**

Statistical parameters:
Since there are 4 classes, n=4
df = n – 1
= 4 – 1
=3
The level of significance or probability level (p) is 0.05, corresponding with 95% confidence limits.
Using chi-squared tables, for df = 3 and p = 0.05, the critical value for χ^2 = 7.81.

Interpreting the result:
The calculated χ^2 value (0.0356) is less than the critical value (7.81).
Therefore, the null hypothesis is accepted at the 0.05 level of significance
Therefore the ratio is 9:3:3:1 and any deviation from this theoretical Mendelian ratio is due to chance.

Conclusion:
The phenotype of the sample is determined by two alleles each of two unlinked genes, where "Purple (P)" is dominant over "yellow (p)" and "Smooth (S)" is dominant over "wrinkled (s)".
The sample is an F_2 generation, produced by self-crossing heterozygous F_1 maize plants (PpSs X PpSs). Inheritance is Mendelian.

TRY THESE QUESTIONS

Question 1

6 In fruit flies, *Drosophila melanogaster*, the allele for grey body colour is dominant to lack body colour, while the allele for normal wing shape is dominant to bent wing shape. A cross was made between a fly heterozygous for both grey body colour and normal wing shape and a fly with black body and bent wing shape. The numbers and phenotypes of the offspring were as follows:

grey body and normal wing shape 83

black body and normal wing shape 85

grey body and bent wing shape 77

black body and bent wing shape 75

(i) Draw a genetic diagram using the following symbols to represent the alleles:

G = grey body colour **g = black body colour**
N = normal wing shape **n = bent wing shape.** [5]

A χ^2 test was then carried out to find out if the results in **(i)** showed that independent assortment had taken place.

(ii) Complete **Table 6.1.**

Table 6.1

offspring	observed numbers	expected numbers
grey body, normal wing shape	83	
black body, normal wing shape	85	
grey body, bent wing shape	77	
black body, bent wing shape	75	
Total	320	320

[1]

(iii) Calculate the χ^2 value for the above data. [2]

(iv) The critical value of χ^2 for this type of investigation with three degrees of freedom is 7.82.
Explain whether the observed results were significantly different from the expected results. [2]

Ref: ZIMSEC June P2 2016 (9190/2 J2016)

Question 2

This practical investigates the dihybrid cross as shown in the photo below of an ear (cob) of corn.

Four different phenotypes, labelled A, B, C and D on the photo, are identified in the ear of corn as follows:

A. Purple and smooth
B. Purple and shrunken
C. Yellow and smooth
D. Yellow and shrunken.

These four grain phenotypes are produced by the following two pairs of heterozygous genes (P & p and S & s) located on two pairs of homologous chromosomes (each gene located on a separate chromosome):

Dominant allele	Recessive allele
P=Purple	p=purple
S=Smooth	s=shrunken

(a) The ear of corn was produced by a dihybrid cross between two heterozygous parents (PpSs X PpSs).
  (i) Use a Punnett square to predict the EXPECTED genotypes and phenotypes of this cross.
  (ii) State the genotypic ratio.
  (iii)State the expected phenotypic ratio. What name is given to this ratio?

(b) An ear of corn has a total of 381 grains which include 216 Purple and smooth, 79 Purple and shrunken, 65 Yellow and smooth, and 21 Yellow and shrunken.
  (i) You formulate a null hypothesis which says: This ear of corn was produced by a dihybrid cross involving two pairs of heterozygous genes (PpSs X PpSs) resulting in a theoretical (expected) ratio of 9:3:3:1.
  Test your hypothesis using the chi-squared test and probability values.
  (ii) What conclusion can you draw from the results of this chi-squared test?

Question 3

Two pure breeding strains of snap-dragon, a garden plant, were obtained. One strain had red flowers and the other had white flowers. The two strains were crossed yielding F_1 plants with pink flowers. The plants were then interbred to produce F_2 plants with the following petal colours.

Red: 62
Pink: 131
White: 67.

The following hypothesis was proposed:
Flower colour is controlled by a single gene with two codominant alleles.

(a) Complete the genetic diagram, using the symbols given to represent the alleles, to explain this cross.

C^r = red; C^W = white

Parental phenotypes	Red flowers	×	White flowers	
Parental genotypes	________		________	
Gametes	________		________	[1]
F_1 genotypes	____________________			
F_1 phenotypes	____________________			
Gametes	____________________			[1]
F_2 genotypes	____________________			
F_2 phenotypes	____________________			
Expected F_2 phenotype ratio	____________________			[1]

(b) A chi-squared (χ^2) test is carried out on the experimental data to determine whether the hypothesis is supported.

(i) Complete **Table 2** by calculating the expected numbers.

Table 2

F2 phenotype	Observed	Expected numbers
Red	62	65
Pink	131	
White	67	
Total	260	260

The χ^2 statistic is calculated in the following way

$$\chi^2 = \sum \frac{(\textbf{Observed} - \textbf{Expected})^2}{\textbf{Expected}}$$

(ii) Calculate the χ^2 value for the above data. Show your working.

[2]

(iii) The critical value of x2 for this type of investigation with two degrees of freedom is 5.991.

Explain whether your answer to (b)(ii) supports the hypothesis.

____

____

____

____ [1]

[TOTAL:8]

Ref: ZIMSEC 9190/2 N2011

SEX LINKAGE

These notes summarize:
• the inheritance of sex-linked genes which are carried on the X-chromosome;
• examples of X-linked inheritance in Drosophila (fruit fly), humans, birds and other animals;
• the inheritance of sex-linked genes which are carried on the Y-chromosome.

Definitions

- **Sex-chromosomes:** The pair of chromosomes in each nucleus that carry the sex-determining genes.
- **Autosomes:** The pairs of chromosomes in each nucleus which do not carry sex-determining genes.
- **Sex linkage:** Refers to genetically controlled characters (but not sex) that are only expressed in one sex (the male). Such characters are never or rarely expressed in the other sex (the female).
- **X-linkage:** When the allele for the linked character is situated only on the X-chromosome.
- **Y-linkage:** When the allele for the linked character is situated only on the Y-chromosome.
- **Gene locus:** The specific position on a chromosome where the allele of a gene is situated. A pair of homologous chromosomes will thus contain two loci (and so two alleles) per gene. The alleles of the gene are at the same position on the homologous chromosomes (see the diagram below).

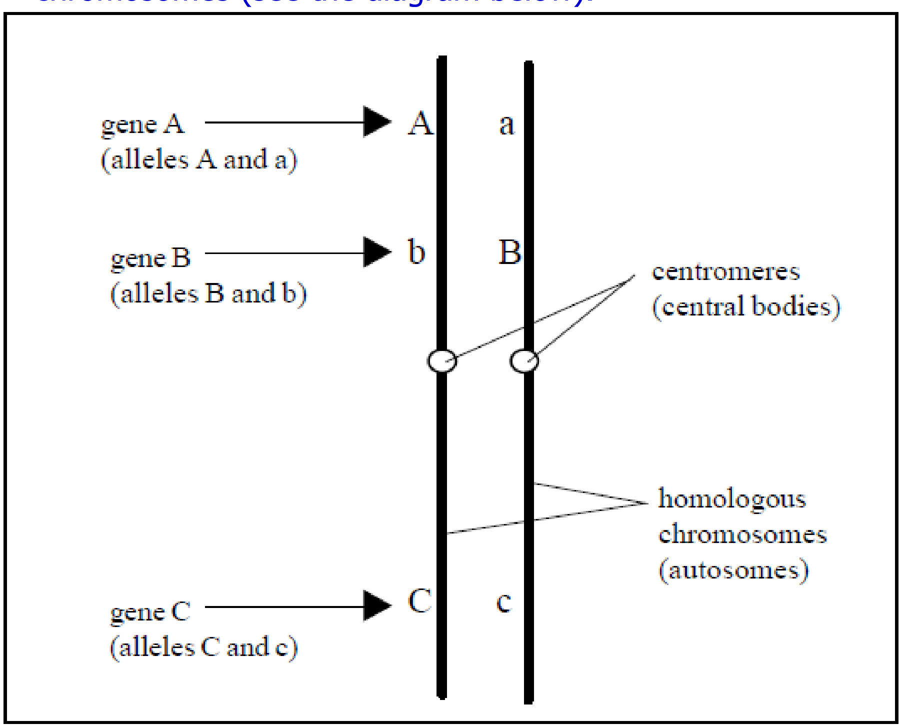
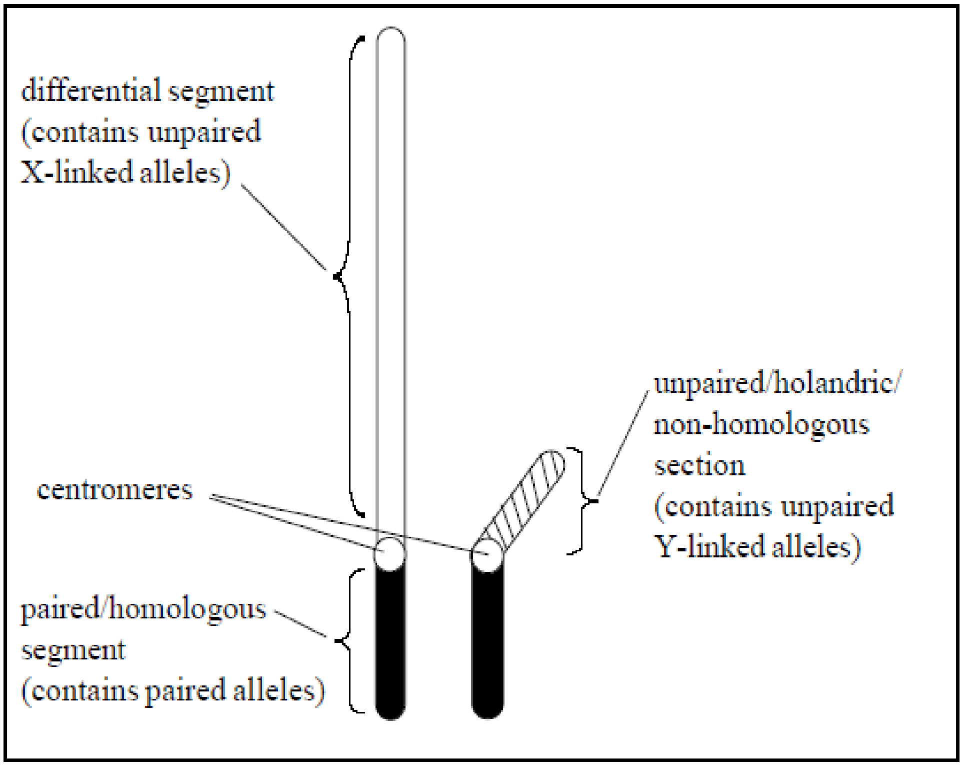


DIAGRAM: A pair of homologous chromosomes showing some gene loci.

Remember - *human nuclei contain 23 pairs of chromosomes, 22 pairs are autosomes and the 23rd pair are sex-chromosomes. In males this pair is designated XY. In females this pair is designated XX.*

The human sex-chromosomes
Note that the chromosomes of an autosomal pair are structurally identical (homologous) to each other and carry identical gene loci. In contrast, the sex-chromosomes are not identical (see diagram below). Thus, although the X-chromosome and Y-chromosome do contain a short similar homologous segment, the X-chromosome is much longer than the Y-chromosome. In addition, the Y-chromosome also contains a short length which is non-homologous to the X- chromosome. This is called the holandric portion.



DIGRAM: Structure of the sex-chromosomes (male)

X-linked inheritance

- The X and Y chromosomes carry the alleles responsible for the development of sex. However other non-sex related alleles are also carried on the sex chromosomes, particularly on the differential segment of the X-chromosome.
- If these non-sex related alleles are recessive then the following occurs;
  - In the male (XY) the recessive allele on the **differential segment of the X-chromosome**, (see diagram above), will be expressed because there is no second X-chromosome with a possible dominant allele present.
  - In the female (XX), because two X-chromosomes are present, a recessive allele on the differential segment of the X-chromosome will usually be masked by the presence of the dominant allele on the other X-chromosome. The recessive allele will only be expressed if both X-chromosomes carry it - that is when the double recessive state is present. A female who is heterozygous for the character is called a carrier, because although she does not exhibit the recessive character she can pass it on to her offspring.

***Remember**:– the female (XX) is referred to as the **homogametic** sex because with respect to sex-chromosomes only one type of gamete (X egg cell) is produced. The male (XY) is referred to as the **heterogametic** sex because with respect to the sex chromosomes two types of gamete are produced (X- sperm and Y-sperm)*

- The diagram that follows illustrates possible male and female genotypes for sex linkage condition leading to **red-green colour-blindness**.

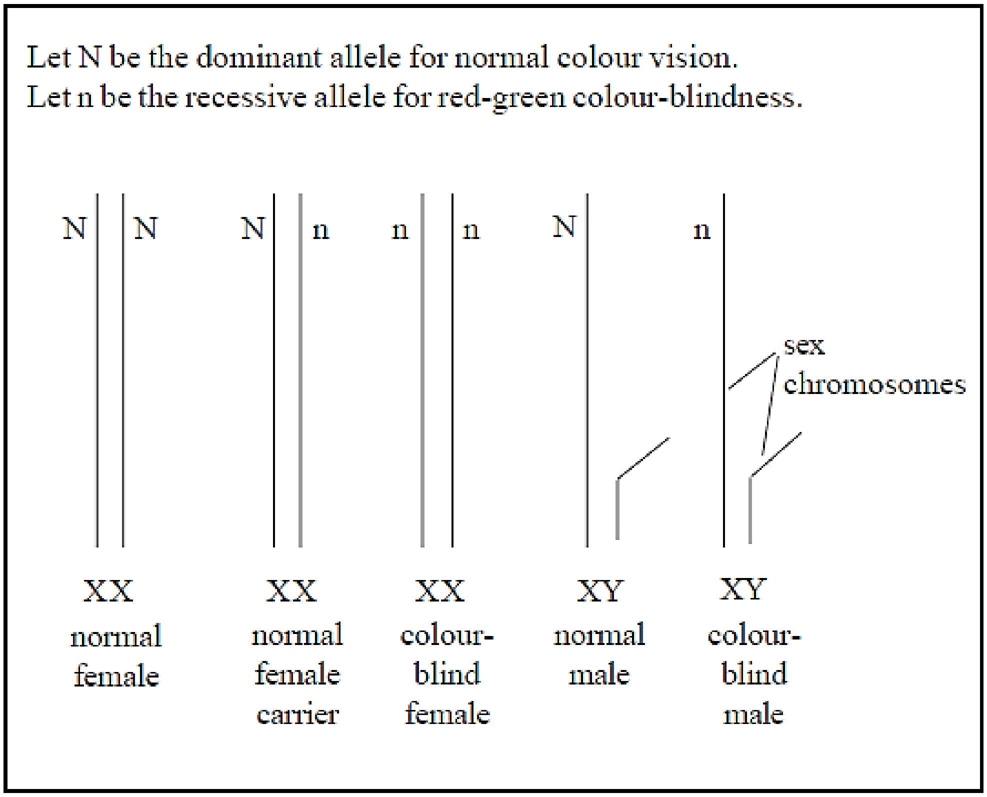


DIAGRAM: Possible male and female genotypes for sex linkage (red green colour-blindness)

- Thus the possible genotypes and phenotypes are;
  - $X^N X^N$ female with normal vision (homozygous dominant).
  - $X^N X^n$ female with normal vision (heterozygous carrier of colour-blindness).
  - $X^n X^n$ female with **red-green colour-blindness** (homozygous recessive).
  - X^N Y male with normal vision.
  - X^n Y male with **red-green colour-blindness**.
- About 2.5% of men suffer from **red-green colour-blindness** but less than 0.1% of women have the condition.
- To have a chance of producing a **red-green colour-blind** female ($X^n X^n$), a **red-green colour-blind** man (X^n Y) must have children by a carrier female ($X^N X^n$). This is illustrated in the cross below.

***Exam Hint** – if you have to answer questions on sex linkage using a genetic cross, then write the cross out in the way shown below, which gives all the information. Alternatively, it is acceptable to use a Punnett square diagram.*

Parental phenotypes:	carrier female	×	colour-blind male	
Parental genotypes:	$X^N X^n$		$X^n Y$	
Gametes:	50% X^N	50% X^n	50% X^n	50% Y
Children:	$X^N X^n$ carrier female	$X^n X^n$ colour-blind female	$X^N Y$ normal male	$X^n Y$ colour-blind male
Ratio:	1 :	1 :	1 :	1

Task: Use a Punnett square to show the above cross.

❖ **Other examples of X-linked inheritance**

- haemophilia A and haemophilia B
- brown enamel on the teeth
- non-functional sweat glands
- absence of central incisors
- some types of deafness
- white forelocks
- juvenile glaucoma
- glucose-6-phosphate dehydrogenase (G6PD) deficiency
- juvenile muscular dystrophy
- retinitis pigmentosa
- coat colour in cats
- white eyes in Drosophila (fruit flies)
- yellow body colour in Drosophila

***Exam Hint** – data questions have been set on several of these conditions. Do not panic, just remember that the inheritance pattern for X-linked inheritance is always the same as for colour blindness.*

Haemophilia

- Haemophilia is a blood disease in which the blood fails to clot adequately.
- Sufferers may bleed to death even from a trivial wound.
- Haemophilia A is most common and is due to a single recessive allele which results in a reduction in the amount of a specific clotting protein called 'anti-haemophiliac factor' (AHF) in the blood. This is the type of haemophilia which was present in Queen Victoria's family and Queen Victoria herself was a carrier of the condition.
- Haemophilia B is rarer and is called 'Christmas disease' after its discoverer. In this condition the recessive allele causes a reduction in the amount of the clotting protein 'plasma thromboplastin component' (PTC) in the blood.
- The following genetic cross shows the possible genotypes of Queen Victoria's children:

Let H be the dominant allele for normal blood clotting ability.
Let h be the recessive allele for haemophilia.

	Queen Victoria				Prince Albert		
Parental phenotypes:	carrier female		×		normal male		
Parental genotypes:	$X^H X^h$		↓		$X^H Y$		
Gametes:	50% X^H	50% X^h		50% X^H	50% Y		
Children:	$X^H X^H$	$X^H X^h$		$X^H Y$	$X^h Y$		
	Normal female	Carrier female		Normal male	Haemophiliac male		
Ratio:	1 :	1 :		1 :	1		

❖ Haemophiliac females can only result from interbreeding between a carrier (or haemophiliac) female and a haemophiliac male. This was impossible until recent years when clotting factors became available to curtail bleeding in haemophiliacs. Before that, haemophiliac males usually died, from haemorrhaging, in childhood.

TASK: Use a genetic cross to shows the possible genotypes between a carrier female and a haemophiliac male.

Remember: *– because haemophilia was normally a lethal condition that killed sufferers before they could reproduce and pass on the haemophilia allele, one would expect that the condition would rapidly become extinct.*
This does not happen because the normal blood clotting alleles constantly mutate into the haemophilia-causing alleles. However, the mutation rate is relatively low, for example, 1 in 20 000 live births in London (haemophilia A).

Thomas Hunt Morgan and the Drosophila (fruit fly)

- Sex linkage was first clearly recognised in 1910 by Morgan and his coworkers using Drosophila (fruit fly).
- Morgan noticed a white-eyed male (mutant) which had suddenly appeared among his cultures of red-eyed flies. When he mated this white-eyed male with its red-eyed sisters he found that all the F1 flies had red eyes. Thus white-eyes was recessive to red-eyes.
- When the F1 flies were allowed to interbreed red-eyed and white-eyed flies appeared in a 3:1 ratio in the F2 generation *but all the white-eyed flies were male*.

❖ Morgan realised that the alleles controlling eye colour must be situated on the differential (unpaired) segment of the X-chromosome:

Let R be the dominant allele for red eyes.
Let r be the recessive allele for white eyes.

Parental phenotypes: Red-eyed female ♀ x White-eyed male ♂

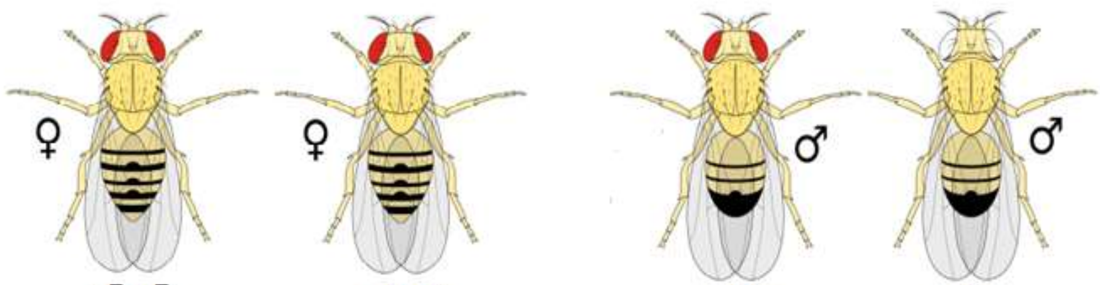
Parental genotypes: X^RX^R X^rY

Gametes: 100% X^R 50% X^r 50% Y

F_1 X^RX^r x X^RY

Red-eyed carrier female ♀ Red-eyed male ♂

Gametes: 50% X^R 50% X^r 50% X^R 50% Y



F_2 X^RX^R Female; X^RX^r Carrier female; X^RY Male; X^rY Male

3 Red-eyed 1 white-eyed

X-linked inheritance in birds

In birds (and also in butterflies and moths) the females are the heterogametic sex (XY) and the males are the homogametic sex (XX). The same principles apply as in the examples of X-linkage above but remember that the pattern of inheritance will be reversed in the sexes, so that the females will tend to show the linked character most frequently and the males will be the carriers.

Examples of X-linked inheritance characters in birds are:

- certain plumage colours in budgerigars (albino, cinnamon, lutino, opaline).
- red and white plumage colours in some breeds of poultry.
- barred plumage in chickens.

To distinguish the sex chromosomes of birds from the X –Y system the symbols Z and W are often used to identify the sex chromosomes. ZZ is male and ZW is female.

Y-linked inheritance

It is incorrect to say that the Y-chromosome is 'genetically empty' although it is true to say that the Y-chromosome contains few alleles. There are very few known examples of Y-linked conditions due to alleles carried on the holandric/non-homologous section of the Y-chromosome. Examples that are suspected to be Y-linked are:

- porcupine skin
- webbed toes
- hairy external ears
- some forms of male sexual dysfunction

These conditions can only occur in men since women do not possess Y chromosomes. They could only be passed from father to son.

Try this question

Red green colour blindness is a sex linked recessive trait in humans. A colour blind man marries a woman with normal vision, whose father is colour blind. **N** is the allele for normal vision, and **n** is the allele for colour blindness.

(a) By means of appropriate symbols, state the genotype of the woman. Explain your answer.

Woman's genotype ____ [1]
Explanation ____
____
____ [2]

(b) (i) Construct a genetic diagram to show the genotypes and phenotypic proportions produced by this cross. [4]

(ii) Calculate the chances (probability) that the first child from this marriage will be a colour blind boy. [2]

Ref. #7ScA. 9190/2 J2006

8.10 TOPIC 2 INHERITED CHANGE AND EVOLUTION

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (ATTITUDES, SKILLS AND KNOWLEDGE)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.10.1 Natural selection	• explain, with examples, how mutations and environment may affect phenotype • explain, with examples, how environmental factors act as forces of natural selection • explain how natural selection may bring about evolution	- Natural Selection - Mutations - Natural selection - Environmental factors - Evolution	• Discussing how mutations and environment may affect phenotype. • Discussing with examples how environmental factors act as forces of natural selection. • Researching and presenting on how natural selection may bring about evolution.	• Print media • ICT tools • Braille software/Jaws
8.10.2 Artificial selection	• describe one example of artificial selection	- Artificial selection	• Outlining the examples of artificial selection.	• ICT tools • Print media • Braille software/Jaws